

ACHIEVE!



maths

**DATA, STATISTICS
& DRAWING
GRAPHS**

Paul Steenson



Made for the
Australian Curriculum





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& DRAWING
GRAPHS**

Paul Steenson

This book was commissioned by Jennifer Gough.
Project management was by Piper Print Media, typesetting
by digiXL and cover design by Gail McManus Graphics.

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ISBN 978 0 99542 757 0

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Introduction

The *Achieve!* series of Maths titles has been developed for teachers who require modified, curriculum-focused activities for secondary school students. It is aimed at students who, because of poor literacy skills, struggle to read and comprehend regular classroom materials. Many of these students did not master the basics of a topic when it was first introduced. Therefore, some activity sheets in this book revisit content that would usually be covered in Year 5 or 6. Some activities are more challenging than others. The activity sheets are also suitable as supplementary material for regular classes. They may also provide useful revision in preparation for state-wide or national numeracy testing.

This book is divided into six units, each of which focuses on a topic or set of skills to do with data, statistics or drawing graphs. The CD-ROM contains all the pages that are featured in the book, as well as a file of illustrations that can be used for other activity sheets. These documents can be easily edited to suit the individual needs of students. Note that photocopying and printing copies from the CD-ROM may change the size of diagrams and graphs.

Teachers can use the material in a variety of ways. For example, a unit could be taught in sequential lessons with a whole class or with students in groups of varying sizes. Alternatively, one or more activity sheets can be used as support material, for individual homework, assessment or revision. Teachers will need to select carefully if individual activity sheets are used as stand-alone tasks, as some activities build on knowledge gained earlier in the unit.

The Teacher's notes provide valuable guidance about using each unit, and contain the following elements.

Objectives – These are the key skills and knowledge learnt through using the activity sheets.

Prior knowledge – This refers to the skills and knowledge that are required for students to complete the tasks.

Background – This section provides a range of different types of useful information to help teach this unit. It can include explanations of key concepts and terminology or further background notes about the subject matter.

Starter activity – Warm-up activities are provided to put the content into proper context and to gain student interest and attention.

Activity sheets and answers –

A summary of the content of each activity sheet is provided, along with teaching tips and selected answers to assist with time-efficient marking.

Extension ideas – Additional activities or discussion ideas are suggested to recap on the main skills, knowledge or terminology taught, as well as ways to extend, assess or consolidate student understanding.

Australian Curriculum alignment

See inside front cover

Teacher's notes

OBJECTIVES

- Understand commonly used language of chance.
- List the possible outcomes of simple events.
- Solve probability problems involving simple events.
- Determine probabilities of events arising from rolling a pair of dice.
- Use tree diagrams to list the possible outcomes of multi-stage experiments.

PRIOR KNOWLEDGE

Students should understand the basic concept of chance. They should be able to simplify a fraction using mental arithmetic or a calculator.

They should understand the meaning of terms like *odd*, *even*, *less than*, *greater than*, *at least*, *vowel* and *consonant*.

BACKGROUND

The early study of probability was associated with gambling and games of chance. Today, it figures significantly in the fields of science, medicine, business and government.

A simple event is an event in which each possible outcome is equally likely. For example, tossing an unbiased coin or selecting a marble at random from a bag.

The problems in these activity sheets are based on equally likely outcomes. A discussion of equally likely outcomes may be a useful starter activity prior to using activity sheet 1.2 Numerical probability.

STARTER ACTIVITY

The class could discuss the likelihood of various familiar events and classify them as impossible, unlikely, even chance, likely or certain to occur. They could also discuss other commonly used words such as *equal chance*, *50-50*, *rarely*, *probable* and *doubtful*, and match them with the terms used in activity sheet 1.1 The language of chance.

ACTIVITY SHEETS

1.1 The language of chance

In this activity sheet, students order commonly used words on a number line between zero (impossible) and one (certain). They are also asked to match these terms to various events.

Answers

- certain: G; impossible: A; even chance: D; unlikely: C; likely: E; highly unlikely: B; very likely: F.
- a very likely b even chance c certain
d impossible e even chance
f highly unlikely g highly unlikely
h very likely

1.2 Numerical probability

In this activity sheet, students assign numerical values to the likelihood of simple events, based on equally likely outcomes.

Answers

- a $\frac{1}{2}$ b $\frac{3}{10}$ c $\frac{1}{5}$ d 0
- a $\frac{1}{6}$ b $\frac{1}{6}$ c $\frac{1}{3}$ d $\frac{1}{2}$ e $\frac{1}{3}$ f $\frac{2}{3}$
- a $\frac{1}{10}$ b $\frac{1}{10}$ c $\frac{9}{10}$ d $\frac{1}{2}$ e $\frac{3}{10}$ f $\frac{3}{10}$
- a $\frac{1}{5}$ b $\frac{1}{5}$ c $\frac{2}{5}$ d $\frac{2}{5}$ e $\frac{2}{5}$ f $\frac{3}{5}$
- a $\frac{7}{15}$ b $\frac{2}{15}$ c $\frac{4}{15}$ d $\frac{2}{15}$ e $\frac{11}{15}$ f $\frac{2}{5}$ g $\frac{11}{15}$ h $\frac{13}{15}$
- a $\frac{1}{6}$ b $\frac{1}{6}$ c $\frac{1}{3}$ d $\frac{2}{3}$ e $\frac{1}{6}$ f $\frac{5}{6}$
- a $\frac{3}{8}$ b $\frac{1}{4}$ c $\frac{1}{4}$ d $\frac{1}{8}$ e $\frac{7}{8}$ f $\frac{5}{8}$ g $\frac{1}{2}$ h $\frac{3}{8}$

Teacher's notes

CONTINUED

1.3 A pair of dice

In this activity sheet, students use 6×6 grids to identify the possible outcomes when a pair of dice is rolled. They use these grids to answer probability questions.

Answers

1

1, 4	2, 4	3, 4	4, 4	5, 4	6, 4
1, 5	2, 5	3, 5	4, 5	5, 5	6, 5
1, 6	2, 6	3, 6	4, 6	5, 6	6, 6

2 a $\frac{1}{36}$ b $\frac{1}{36}$ c $\frac{1}{6}$ d $\frac{1}{18}$ e $\frac{11}{36}$ f $\frac{11}{36}$ g $\frac{5}{18}$ h $\frac{1}{4}$

3 a

Total result	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

b 7 c 4 d 2 and 12

4 a $\frac{1}{6}$ b $\frac{1}{9}$ c $\frac{1}{36}$ d $\frac{1}{6}$ e $\frac{1}{6}$ f $\frac{5}{36}$

5 a $\frac{1}{2}$ b $\frac{1}{2}$ c $\frac{2}{9}$

6 A

1.4 Tree diagrams

In this activity sheet, students use tree diagrams to list the possible outcomes of multi-stage experiments and answer associated probability questions.

Answers

1 a $\frac{1}{4}$ b $\frac{1}{2}$

2 a $\frac{1}{8}$ b $\frac{1}{8}$ c $\frac{1}{2}$ d $\frac{1}{2}$ e $\frac{1}{4}$ f $\frac{3}{8}$ g $\frac{1}{8}$

3 a $\frac{1}{6}$ b $\frac{1}{6}$ c $\frac{1}{3}$ d $\frac{1}{3}$ e $\frac{1}{3}$

4 a $\frac{1}{9}$ b $\frac{1}{3}$ c $\frac{2}{3}$ d $\frac{1}{9}$ e $\frac{2}{9}$ f $\frac{5}{9}$

EXTENSION IDEAS

Students could investigate probabilities arising from simple board or card games.

1.1 The language of chance

Kaitlyn and Eli were planning to go to the beach.

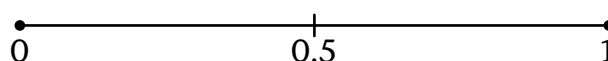
We use words like *chance*, *most likely*, *probably*, *unlikely* and *certain* in our everyday conversations. These words all have meaning in the study of chance and probability.

Probability is a way of describing the **chance** of something happening.

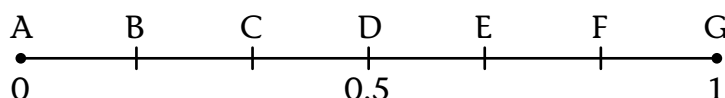
An event that **will not happen** has a probability of **zero (0)**.

An event that **will definitely occur** has a probability of **1**.

These are the two end points of the probability line.



- 1 Match the probability terms with the positions marked by the letters on the probability line below.



Certain ____ Even chance ____ Likely ____ Very likely ____
 Impossible ____ Unlikely ____ Highly unlikely ____

- 2 Choose the term that best describes each event.

- | | | | |
|---------------|---------------|-------------------|--------------|
| • certain | • likely | • unlikely | • impossible |
| • very likely | • even chance | • highly unlikely | |

- a Finding sand on a beach _____
 b A coin toss landing heads up _____
 c The sun rising in the morning _____
 d A wombat flying across the playground _____
 e A newborn child being a boy _____
 f Rolling a double 6 with a pair of normal dice _____
 g Being struck by lightning today _____
 h A student at your school having a birthday today _____

1.2 Numerical probability

If each possible outcome is equally likely, then:

$$\text{The probability of an event} = \frac{\text{Number of ways the event can occur}}{\text{Number of all possible outcomes}}$$

The possible outcomes are the different results (or events) that can happen.

Most probability questions involve **random** events.

A selection is made at random when the objects are the same size and the selection is made without looking. When a selection is made at random, each outcome is equally likely.

Probabilities are usually written as fractions. They can also be written as decimals or percentages. Always simplify the answer where possible.

EXAMPLE

A bag contains five marbles numbered 1, 2, 3, 4, and 5.

The possible outcomes are: 1, 2, 3, 4, 5. There are 5 possible outcomes

One marble is selected at random. What is the probability that it is numbered 5?

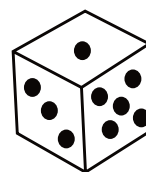
Only 1 marble is numbered 5.

So the probability is $\frac{\text{Number of ways the event can occur}}{\text{Number of all possible outcomes}} = \frac{1}{5}$

- 1 A box contains 10 coloured balls. 5 balls are green, 3 balls are red and 2 balls are yellow. One ball is selected at random. What is the probability that it is:

a green? _____ **b** red? _____ **c** yellow? _____ **d** blue? _____

- 2 A single die is thrown.
What is the probability the result is:



a 6? _____ **b** 2? _____
c 4 or 5? _____ **d** an even number? _____
e a number less than 3? _____ **f** a number greater than 2? _____

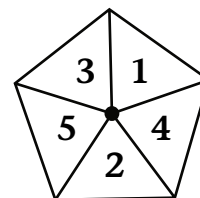
- 3 Ten identical cards are numbered 1 to 10. The cards are shuffled. One card is selected at random. What is the probability that the number written on it is:

a 10? _____ **b** 5? _____
c not 10? _____ **d** an odd number? _____
e a number less than 4? _____ **f** a number greater than 7? _____

1.2 Numerical probability

CONTINUED

- 4** A spinner is made using a regular pentagon. A regular pentagon has five equal sides. The equal sections are labelled with the numbers 1, 2, 3, 4 and 5.

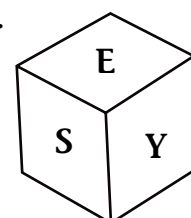


What is the probability that it will give the result:

- a** 4? _____ **b** 3? _____
c a number less than 3? _____ **d** a number greater than 3? _____
e an even number? _____ **f** an odd number? _____
- 5** A bag contains 15 marbles. 7 marbles are blue, 4 marbles are red, 2 marbles are green and 2 marbles are yellow. A marble is taken from the bag at random. What is the probability that it is:

- a** blue? _____ **b** yellow? _____
c red? _____ **d** green? _____
e either blue or red? _____ **f** either red or yellow? _____
g blue, green or yellow? _____ **h** not yellow? _____

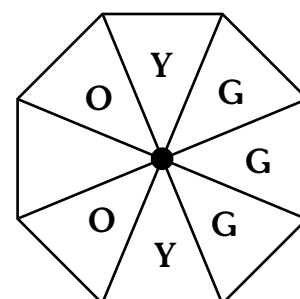
- 6** The letters of the word SYDNEY were stuck on the faces of a die. Craig rolled this die.



What is the probability that the result is:

- a** S? _____ **b** D? _____
c Y? _____ **d** not Y? _____
e a vowel? _____ **f** a consonant? _____

- 7** Monique made a spinner using a regular octagon. A regular octagon has 8 equal sides. She coloured three of the sections green. Two sections were yellow and two sections were orange. The remaining section was white.



What is the probability that it will give the result:

- a** green? _____ **b** yellow? _____
c orange? _____ **d** white? _____
e not white? _____ **f** not green? _____
g green or white? _____ **h** white or yellow? _____

1.3 A pair of dice

Meghan and Sonia are playing a board game using a pair of dice. Sonia needs to throw a double 6 to win.

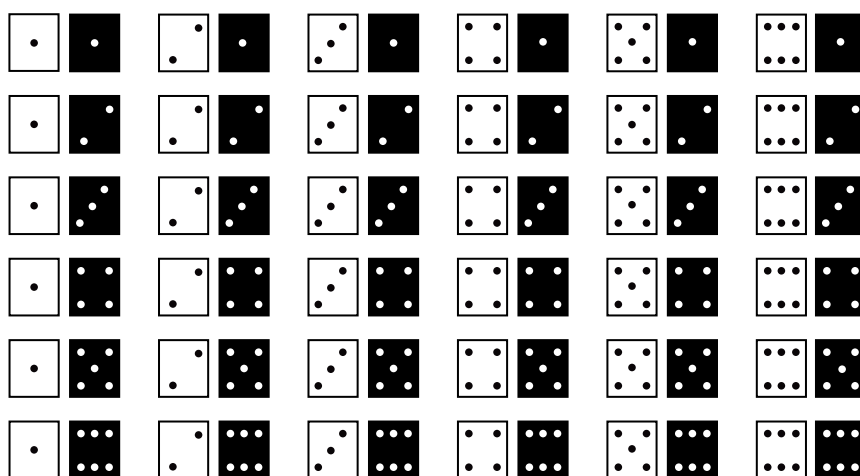
What is the probability that Sonia will get a double 6 on her next throw?

When you throw a pair of normal dice, each die acts independently.



Think of the dice as having different colours, white and black.

Each of the six results on the white die can be matched with any of the six results on the black die. There are 36 different pairs. They are shown in this diagram.



Each of these results is equally likely. So, the probability of a double 6 is $\frac{1}{36}$. The probabilities do *not* change if the dice are the same colour.

- 1** The results of throwing two dice can be shown in a table of ordered pairs. Complete this table.

1, 1	2, 1	3, 1	4, 1	5, 1	6, 1
1, 2	2, 2	3, 2	4, 2	5, 2	6, 2
1, 3	2, 3	3, 3	4, 3	5, 3	6, 3
1, 4					
1, 5					
1, 6					

- 2** A pair of dice is rolled. What is the probability that:

- a** the result is double 5? _____ **b** the result is double 1? _____
c the result is any double? _____ **d** 1 and 2 are showing? _____
e at least one die shows 6? _____ **f** at least one die shows 4? _____
g only one die shows 6? _____ **h** both numbers are even? _____

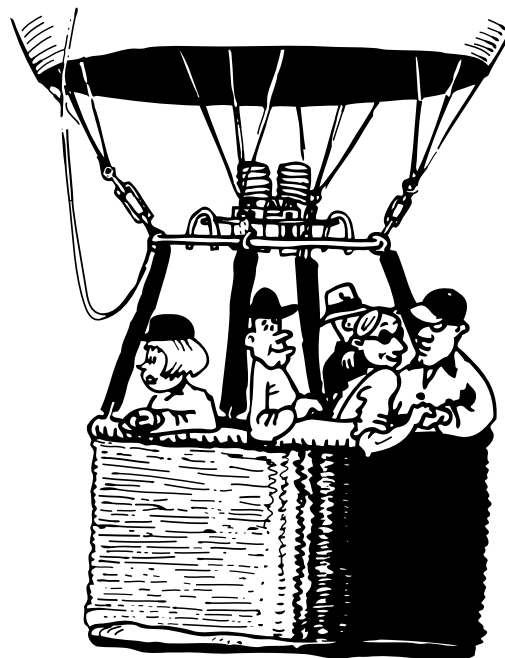
1.3 A pair of dice

CONTINUED

3 The following table shows the **total result** when a pair of dice is thrown.

a Complete the missing numbers.

		Total result					
		1	2	3	4	5	6
1	2	2	3	4	5	6	7
2	3	3	4				
3	4	4					
4	5	5					
5	6	6	7				
6	7	7	8	9	10	11	12



b What is the most common total? _____

c In how many ways can a total of 5 occur? _____

d What are the two least likely totals? _____

4 A pair of dice is rolled. What is the probability of:

a a total of 7? _____

b a total of 9? _____

c a total of 12? _____

d a total more than 9? _____

e a total less than 5? _____

f a total between 9 and 12? _____

5 A pair of dice is rolled. What is the probability that:

a the total is an even number? _____

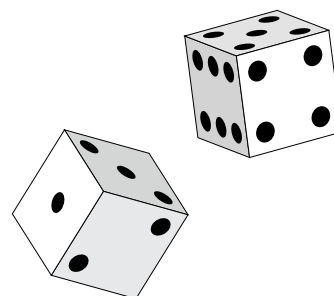
b the total is an odd number? _____

c the total score is 7 or 11? _____

6 Which has the greater probability – (A) or (B)?

(A) Rolling a 6 with 1 die

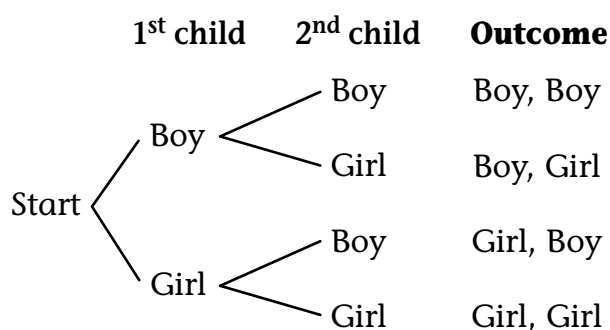
(B) Getting a total of 6 when a pair of dice is rolled _____



1.4 Tree diagrams

EXAMPLE

Liam has a brother but does not have a sister.
He wanted to know the probability that in a family of 2 children, both children are boys.
A new child is equally likely to be a boy or a girl.
This is the same for both the first and second child.
This tree diagram shows the possible outcomes.

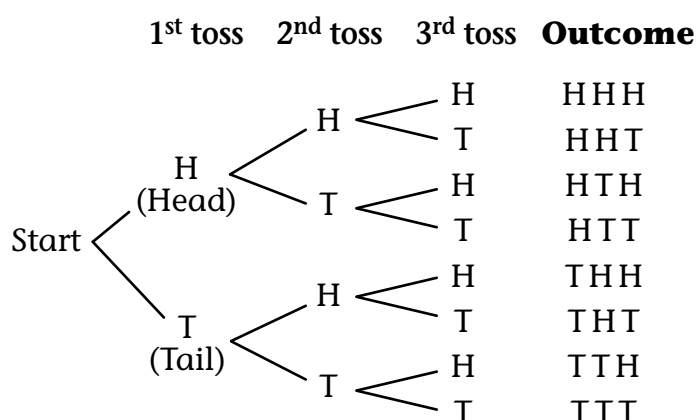


Each outcome is equally likely.
So, the probability of two boys is $\frac{1}{4}$.



- 1 In a family of two children, what is the probability that the children are:
a both girls? _____ **b** a boy and a girl (in either order)? _____

- 2 Peta tossed a coin three times.
At each toss of a coin, a head and a tail are equally likely.
This tree diagram shows the possible outcomes.



What is the probability that the result is:

- a** 3 heads? _____ **b** 3 tails? _____
c at least 2 heads? _____ **d** at least 2 tails? _____
e 3 heads or 3 tails? _____
f 2 heads and a tail in any order? _____
g a head, a tail, then a head, in that order? _____

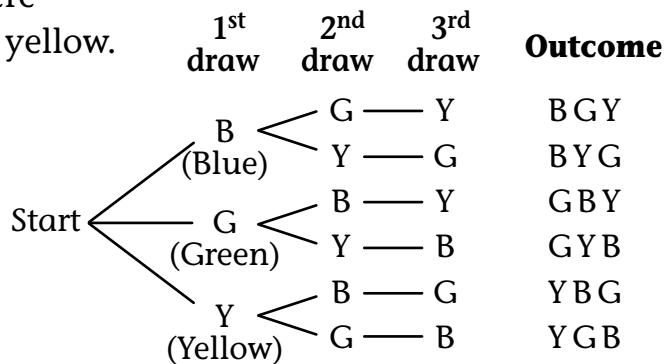
1.4 Tree diagrams

CONTINUED

- 3 Alison had three marbles. They were different colours – blue, green and yellow. She placed the marbles in a bag.

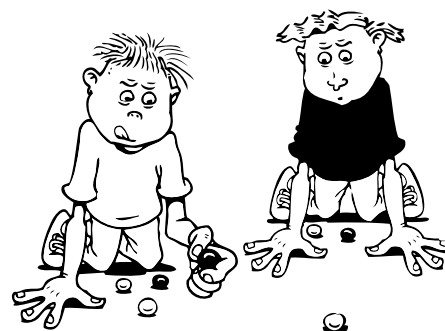
Alison selected the marbles from the bag, one at a time, without looking.

This tree diagram shows the possible outcomes. They are equally likely.



What is the probability that:

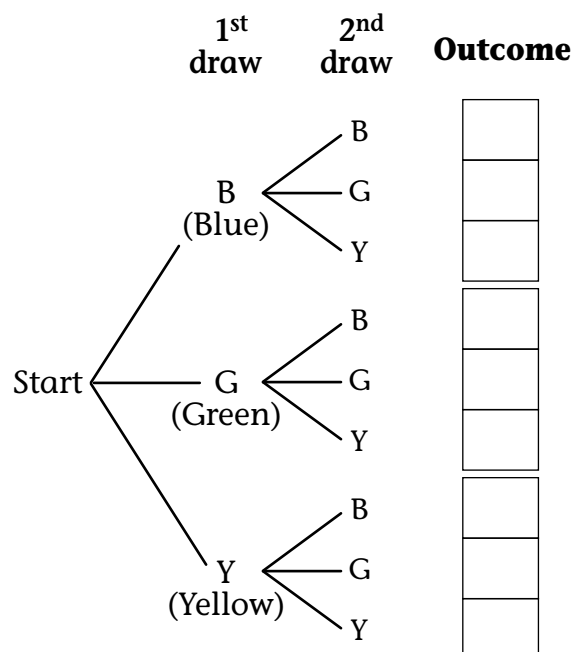
- a the order was blue, green, yellow? _____
- b the order was yellow, blue, green? _____
- c the first marble was blue? _____
- d the first marble was yellow? _____
- e the last marble was green? _____



- 4 Josef had three coloured discs. They were different colours – blue, green and yellow. He placed the discs in a bag.

Josef selected a disc from the bag, without looking. He noted the colour and put the disc back in the bag. Josef then selected a second disc, without looking.

Complete this tree diagram to show the possible outcomes. They are equally likely.



- What is the probability that:
- a both discs are yellow? _____
 - b the two discs are the same colour? _____
 - c the two discs are different colours? _____
 - d he drew the blue disc first, then the green disc? _____
 - e he drew a yellow disc and a blue disc, in either order? _____
 - f at least one disc was yellow? _____

Teacher's notes

OBJECTIVES

- Recognise the difference between categorical and numerical data.
- Recognise the difference between conducting a census and surveying a sample.
- Use a tally to organise data in a frequency table.
- Read and interpret data presented in a frequency table

PRIOR KNOWLEDGE

Students should be familiar with simple displays of data.

BACKGROUND

Students should be aware of the extensive use of data in society. In everyday life, there is a constant need for people to understand, read and interpret information presented in tabular or graphical forms. This unit aims to develop an understanding of how data may be collected, organised and presented.

STARTER ACTIVITY

The teacher could lead a discussion of the various ways students use data in everyday life; for example, most popular TV show, daily rainfall and temperature. Students could be asked to give other examples. These could be listed and the type of data (numerical or categorical) identified as an introduction to Activity sheet 2.1.

ACTIVITY SHEETS

2.1 Types of data

This activity sheet introduces the two main types of data – numerical and categorical data. Students are asked to identify the type of data in various situations. They are also asked to classify numerical data as discrete or continuous.

Answers

- 1 **a** Numerical **b** Categorical **c** Numerical
d Categorical **e** Numerical **f** Numerical
g Categorical **h** Numerical
- 2 **a** Continuous **b** Continuous **c** Discrete
d Continuous **e** Continuous **f** Discrete
g Continuous **h** Discrete

2.2 Collecting data

This activity sheet describes methods for collecting data. Students are asked whether they would conduct a census or survey a sample in various situations.

Answers

- 1 **a** Sample **b** Sample **c** Sample
d Census **e** Sample **f** Sample
g Census **h** Sample **i** Census
j Sample **k** Sample **l** Census
m Census **n** Sample

2.3 Organising data

In this activity, students use tally marks to organise data in a frequency table.

Answers

1

Number of children	1	2	3	4	5	6
Frequency	4	11	6	4	1	1

2 **a**

Colour	White	Grey	Blue	Black	Red	Green	Others
Frequency	18	12	8	7	10	4	6

b 65

3 **a** 20 **b** 13

c

Score	13	14	15	16	17	18	19	20
Frequency	1	3	5	4	7	2	4	1

Teacher's notes

CONTINUED

2.4 Using frequency tables

In this activity sheet, students read and answer questions about data presented in a frequency table.

Answers

- 1 a 100 b Friday c 20 d 80
- 2 a

Age	11	12	13	14	15	16	17	18	19
Frequency	2	7	10	9	12	8	7	6	3

- b 9 c 15 d 16 e 19

2.5 Maximum temperature

In this activity sheet, students organise the data in a frequency table and answer questions about the data.

Answers

- 1 19
- 2 28
- 3

Temperature	19	20	21	22	23	24	25	26	27	28
Frequency	2	3	4	3	1	2	6	4	3	2

- 4 a 1 b 3
- 5 25°C
- 6 a 15 b 5
- 7 13

EXTENSION IDEAS

Students could use computer software such as a spreadsheet to organise data in a table.

2.1 Types of data

Data is the information collected from a survey or census. There are two types of data:

- **Numerical data** consists of numbers. It is collected by counting or measuring. For example: the number of students in each class can be counted; the heights of students in a class can be measured. Numerical data is often displayed in a frequency histogram.
- **Categorical data** consists of words or letters. For example: the colour of each student's hair may be brown, black, blonde or red; your grade in a class test may be A, B, C, D or E. Categorical data is often displayed in a picture graph or a column graph.



1 State whether the following data is Numerical (N) or Categorical (C).

- | | |
|--|--|
| a The number of days in each month ☐ | e The number of children in a family ☐ |
| b Each student's favourite actor ☐ | f The number of pets owned ☐ |
| c The daily temperature ☐ | g The eye colour of students ☐ |
| d The type of car owned ☐ | h The number of marbles in a bag ☐ |

Numerical data can be described as **discrete** or **continuous**.

- **Discrete** data is obtained by counting. For example, the number of children in a family.
- **Continuous** data is obtained by measuring. The results are often rounded off to a required accuracy. For example, the length of telephone calls.

2 State whether the following numerical data is Discrete (D) or Continuous (C).

- | | |
|---|---|
| a The masses of students in a class ☐ | e The amount of time spent using the internet each week ☐ |
| b The maximum temperature each day ☐ | f The number of planes landing at an airport each day ☐ |
| c The number of goals scored by players during a season ☐ | g The heights of trees in the playground ☐ |
| d The heights of the world's 10 tallest buildings ☐ | h The number of cars passing the school each hour ☐ |

2.2 Collecting data

Data can be obtained from many sources. It is available on the internet and in the media. Often you need to conduct a **survey** or a **census** to obtain the information you need.

A **census** is taken when you ask **everyone** to provide the information. Every five years, the Australian Bureau of Statistics conducts a national census. Information is collected from every member of the community.

It is not easy to conduct a census every time you need data. A **survey** is conducted using a **sample** from the population. The size of the sample and how it is selected can affect the accuracy of the results.

There are a number of ways to obtain a sample.



- **Random sampling:** For example, everyone's name is placed in a hat. The names of 20 people are selected without looking.
- **Systematic sampling:** For example, every fifth name on an alphabetical list is selected.
- **Stratified sampling:** For example, equal numbers of girls and boys are selected.

1 Would you **conduct a census** (write C) or **survey a sample** (write S) to:

- | | |
|--|---|
| a find the most popular TV programme last week? ☐ | h determine the most commonly used letter of the English alphabet? ☐ |
| b predict the result of an election? ☐ | i select a new government? ☐ |
| c check the stated average contents on a box of matches? ☐ | j find the most popular breed of pet dog? ☐ |
| d find the average size of an Australian family? ☐ | k predict the number of frozen peas in a 1 kilogram packet? ☐ |
| e test a new food product to check the fat content? ☐ | l find the number of children under 15 years of age in your street? ☐ |
| f determine the most popular song this week? ☐ | m find the percentage of students with blue eyes at your school? ☐ |
| g find how many students at your school have a driver's licence? ☐ | n estimate the average life of a new type of light globe? ☐ |

2.3 Organising data

Suzanne conducted a survey of the students in her class. She asked each student how many children were in their family. She recorded the following data:

When you conduct a survey you can collect a lot of data. You need a way to organise and display the information so that it is easily understood.

Peter: 2	Melanie: 1	Paul: 2
Dianne: 2	John: 4	Jane: 1
Anne: 3	David: 2	Debbie: 3
Stephen: 2	Rebecca: 2	Geoffrey: 4
Alison: 2	Elizabeth: 3	Nick: 2
Maria: 5	Alan: 4	Bradley: 1
Jemma: 2	Gary: 3	Toula: 3
Joanna: 2	Wayne: 6	Laura: 1
Marko: 4	Ahmed: 3	Suzanne: 2

One way to organise data is to use a **frequency table** and **tally marks**.

To complete a frequency table:

- Step 1** List all the different scores or results in the **first column**.
When you are using numerical data, first circle the smallest and largest scores in the data. Then, list the different scores in ascending order (smallest to largest).
- Step 2** Work across each row in the data, and put a tally mark (a stroke) for each score. Every fifth stroke beside a score is made diagonally.
- Step 3** Count up the number of tally marks for each score. Write these numbers in the **frequency column**.
- Step 4** Find the total of the frequency column. Check that this is equal to the number of scores in the original data.
- Step 5** Write a **heading** (or title) for your table.

- 1** Arrange Suzanne's data in this frequency table.



Number of children in the family

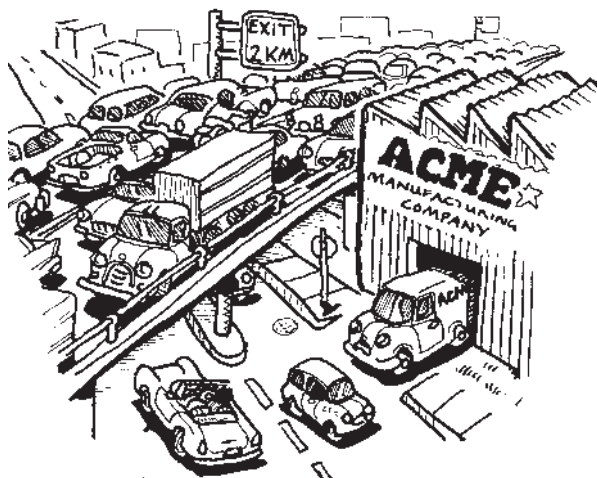
Number of children	Tally	Frequency
1		
2		
3		
4		
5		
6		

2.3 Organising data

CONTINUED

2 Lachlan recorded the colours of cars passing the school. He recorded the data, using tally marks, in this table.

- a Complete the frequency column in this table.
- b How many cars did Lachlan count in the survey? _____



Colours of cars

Colour	Tally	Frequency
White		
Grey		
Blue		
Black		
Red		
Green		
Others		

3 The list at the right shows the scores of 27 students in a science quiz.

17	15	17	18	19	17	16	14	19
13	20	15	15	16	19	16	17	14
15	14	17	18	15	17	19	17	16

- a What was the **highest** score in the quiz?

- b What was the **lowest** score in the quiz?

- c Organise this data in the frequency table.

Scores in science quiz

Score	Tally	Frequency

2.4 Using frequency tables

- 1** Jason conducted a survey at his school.
He asked students: *On which day of the week were you born?*
He organised the data in this frequency table.

Day of birth		
Day	Tally	Frequency
Monday		16
Tuesday		13
Wednesday		17
Thursday		15
Friday		19
Saturday		11
Sunday		9

- a** How many students were counted in the survey?

- b** Which day was the most common day of birth?

- c** How many people were born on a Saturday or Sunday? _____
- d** How many people were born on a weekday (Monday to Friday)? _____

- 2** Christine recorded the ages, in years, of the members of her hockey club.
She organised the data in this frequency table.

Ages of hockey club members

Age	Tally	Frequency
11		
12		
13		
14		
15		
16		
17		
18		
19		

- a** Complete the frequency column in this table.
- b** How many members were 14 years old?

- c** Which age was most common?

- d** How many members were more than 16 years old?

- e** How many members were less than 14 years old?

2.5 Maximum temperatures

The following numbers are the maximum temperatures (in °C) recorded at Woodport on 30 consecutive days.

22	24	25	20	22	26	28	20	21	25
19	25	22	21	21	24	25	26	25	27
26	28	27	26	25	27	20	19	21	23

- 1 What was the lowest maximum temperature recorded? _____
- 2 What was the highest maximum temperature recorded? _____

- 3 Complete the frequency table to organise this data.

- 4 How many days had a maximum temperature of:

- a 23°C? _____
- b 27°C? _____

- 5 Which maximum temperature was recorded most often?

- 6 How many days had a maximum temperature

- a greater than 24°C? _____
- b less than 21°C? _____

- 7 How many days had a maximum temperature between 24°C and 28°C?

Maximum temperature

Temperature (°C)	Tally	Frequency



Teacher's notes

OBJECTIVES

- Identify the range of a set of scores.
- Determine the mode and median for a set of data.
- Calculate the mean for a small set of data.
- Calculate the mean for data presented in a frequency table.
- Use the mean, median and mode to compare two sets of data.

PRIOR KNOWLEDGE

Students should be able to complete the four basic arithmetic operations using a calculator. They should be able to round an answer to the required degree of accuracy.

They should be able to arrange a set of numbers in order from smallest to largest.

They should be able to read and interpret data presented in a frequency table.

BACKGROUND

Students should be aware of the extensive use of data in everyday life. They may have already used the range and the mean in other subjects at school. This unit aims to develop an understanding of the basic methods of analysing data – the range, the mode, the median and the mean.

STARTER ACTIVITY

Ask students to identify the smallest number and largest number in a list. Then ask the students to arrange the numbers in order from smallest to largest.

ACTIVITY SHEETS

3.1 The range

In this activity sheet, students identify the lowest score and the highest score, and then calculate the range of a set of scores.

Answers

1 a 5 b 4 c 5 d 24 e 8

2 9°C

3 a 6 b 19 c Justine

3.2 The mode

In this activity sheet, students determine the mode of a set of scores. In question 3, they are asked to identify the mode when the data is presented in a frequency table.

Answers

1 a 5 b 24 c 12 d 17 e 10

2 a 11, 12 b 24, 26 c 16, 17, 19
d 7, 8 e 24, 26

3 a 3 b 3 min c 4 d 3 min

3.3 The median

In this activity sheet, students identify the median of a set of scores. The questions cover both an odd and even number of scores.

Answers

1 a 4 b 13 c 17 d 55.5

2 a 12, 17, 20, 25, 26; median = 20

b 14, 17, 17, 18, 19, 20; median = 17.5

c 20, 22, 23, 23, 24, 26, 27; median = 23

d 31, 32, 32, 35, 36, 37; median = 33.5

3.4 The mean

In this activity sheet, students use the rule:

mean = the sum of the scores ÷ the number of scores to calculate the mean of a set of scores.

Answers

1 a 305 kg b 61 kg

2 a 312 b 78

3 a 9 b 16 c 53 d 65 e 36

4 164 cm

Teacher's notes

CONTINUED

3.5 Calculating the mean

In this activity sheet, students calculate the mean when the data is organised in a frequency table. This is done by first completing a score $\times$ frequency column in the table. Then they use the rule: **mean = the sum of the score $\times$ frequency column $\div$ the sum of the frequency column** to calculate the mean.

Answers

- 1 a 174, 704, 1 246, 1 080, 910, 276, 93
b 50 c 4 483 d 90
- 2 a 0, 3, 10, 6, 12, 5 b 32 c 36 d 1.125
- 3 a

Score	Frequency	Score $\times$ Frequency
0	3	0
1	8	8
2	14	28
3	10	30
4	3	12
5	2	10

- b Mean = 2.2

3.6 Using the mean, median and mode

In this activity sheet, students use the mean, median and mode to answer questions about given data. They are also asked to compare two or more sets of data using these measures of central tendency.

Answers

- 1 a mean b mode c median d mean
e mode f mean
- 2 a 71 b 72.2 c 72 d 72.1 e Richard
- 3 a Michaela b 46 c 43.67 d Katherine
- 4 a 12 b 15 c 14 d 14 e 15
- 5 a 96.1 mm b 78.6 mm c 79.5 mm
d Brisbane

EXTENSION IDEAS

Students could use a scientific or graphics calculator to determine the mean of a set of scores.

They could discuss situations when it may be more appropriate to use the mode or the median, rather than the mean, when analysing a set of data.

3.1 The range

Maree coaches a netball team. She wants to compare the results of two players. She calculated the **range** of each player's percentages to see who is more consistent.

- The **range** is a **measure** of the spread of a set of scores.
- The range of a set of scores is the highest score subtract the lowest score.

Range = Highest score – Lowest score

EXAMPLE

What is the range of these scores:

75, 70, 71, 70, 78?

Highest score = 78 Lowest score = 70

Range = Highest score – Lowest score
= 78 – 70

Range = 8



- What is the range of each set of scores?
It may help to circle the highest and lowest scores.
 - 11, 14, 16, 13, 15, 11 range = _____
 - 12, 14, 13, 12, 11, 15, 13 range = _____
 - 28, 31, 29, 32, 27, 28, 30, 28, 30 range = _____
 - 67, 63, 59, 45, 67, 63, 69, 47, 63, 60, 55 range = _____
 - 14, 15, 16, 11, 18, 15, 12, 16, 17, 19, 17, 15, 14, 13, 14 range = _____
- The temperatures recorded each hour between 6 am and 12 noon were:
15°C, 16°C, 17°C, 20°C, 21°C, 22°C, 24°C.
What is the range of these temperatures? _____
- Justine and Francesca play in the same netball team.
Their recent percentages of successful shots at goal are shown below.
Justine: 72, 74, 75, 78, 76
Francesca: 70, 72, 76, 88, 69
 - What is the range of Justine's percentages? _____
 - What is the range of Francesca's percentages? _____
 - Which player is more consistent? _____

3.2 The mode

If a score occurs more than any other score, it is called the **mode**.
The **mode** is the score that occurs most often. It is the score with the highest frequency.

EXAMPLE

What is the mode of these scores: 12, 13, 14, 14, 14, 15, 15, 16?
The score 14 occurs more than any other score. The mode is 14.

1 What is the mode of each set of scores?

- a 3, 4, 4, 5, 5, 5, 6, 7 _____
- b 20, 22, 23, 23, 24, 24, 24, 24, 25 _____
- c 12, 11, 12, 12, 13, 12, 14 _____
- d 15, 17, 17, 16, 15, 18, 17, 16 _____
- e 10, 10, 12, 15, 10, 15, 10, 12, 14, 15 _____

Sometimes a set of scores will have **more than one mode**.
When two scores have the highest frequency,
they are both called modes.

2 State the modes of these sets of scores.

- a 10, 11, 11, 12, 12, 13, 14, 15 _____
- b 22, 23, 24, 24, 24, 25, 25, 26, 26, 26 _____
- c 15, 16, 16, 17, 17, 18, 19, 19, 20 _____
- d 4, 5, 7, 6, 7, 8, 10, 8 _____
- e 24, 30, 27, 25, 24, 26, 23, 26, 28 _____

3 Trevor recorded the number of minutes
his train was late over a period of 15 days.
The data is shown in this table.

- a How many times was the train 2 minutes late?

- b Which time was the most common? _____
- c What was the greatest number
of minutes the train was late? _____
- d Which time is the mode? _____

Time (minutes)	Frequency (days)
0	4
1	2
2	3
3	5
4	1



3.3 The median

The **median** is the **middle score** after the scores are arranged in order from smallest to largest.

Half the scores are to the left of the median. The other half are to the right of the median.

The median is a measure of the central tendency of the scores. It is not affected by unusually high or low scores.

EXAMPLE

What is the median of these scores:

12, 15, 16, 16, 17, 18, 20?

The scores are arranged in order.

The middle score is 16.

Three scores are to the right.

Three scores are to the left.

The median is 16.

When there is an *even* number of scores, the median is halfway between the two middle scores. Add the two middle scores then divide by 2.

The median is not always one of the scores.

- 1** Find the median of each set of scores.

a 2, 3, 4, 4, 5, 6, 8

b 11, 11, 12, 13, 14, 15, 16

c 14, 15, 17, 17, 18, 19

d 52, 54, 54, 55, 56, 58 60, 60

EXAMPLE

What is the median of these scores:

23, 25, 26, 27, 29, 29?

The scores are arranged in order.

There is an *even* number of scores (6).

The middle two scores are 26 and 27.

Two scores are larger.

Two scores are smaller.

The median is halfway between 26 and 27.

The median is 26.5.

- 2** Arrange these scores in order, smallest to largest. Then state the median of each set of scores.

a 17, 12, 20, 25, 26

The median is _____

b 14, 17, 20, 17, 18, 19

The median is _____

c 22, 26, 27, 20, 23, 24, 23

The median is _____

d 36, 31, 32, 37, 32, 35

The median is _____

3.4 The mean

Rochelle plays cricket with her school team. Her recent batting scores were 10, 25, 17 and 40 runs. The coach wanted to calculate her average score.

The term **average** is used in many situations. For example, batting averages in cricket or the average monthly rainfall.

The **mean** is the mathematical term for the average of a set of scores.

To calculate the mean we add all the scores and then divide the sum by the number of scores.

Mean = the sum of the scores $\div$ the number of scores

EXAMPLE

Calculate the mean of 10, 25, 17 and 40.

The sum of the scores = $10 + 25 + 17 + 40 = 92$

The number of scores = 4

Mean = the sum of the scores $\div$ the number of scores
 $= 92 \div 4 = 23$

Rochelle's average batting score was 23 runs.



1 The masses of five people are: 73 kg, 68 kg, 49 kg, 61 kg and 54 kg.

a What is their total mass? _____

b What is their mean (average) mass? _____

2 Toni received the following scores in her last 4 tests: 89, 63, 76, 84.

a What is the total of her scores? _____

b What is her mean (average) score in these tests? _____

3 Calculate the mean of each set of scores.

a 8, 9, 6, 12, 10 _____

b 14, 11, 15, 21, 18, 17 _____

c 42, 36, 45, 76, 58, 61 _____

d 72, 84, 68, 46, 54, 76, 55 _____

e 30, 32, 45, 33, 47, 23, 39, 35, 40 _____

4 The heights of eight students, to the nearest centimetre, are:
 168 cm, 150 cm, 171 cm, 175 cm, 161 cm, 155 cm, 158 cm, 173 cm.

Calculate the mean of these heights, to the nearest centimetre. _____

3.5 Calculating the mean

Douglas randomly selected 50 boxes of toothpicks. He counted the number of toothpicks in each box. The results are recorded in the following frequency table.

Douglas wanted to calculate the mean (average) number of toothpicks in these boxes.

To calculate the mean when the data is presented in a frequency table:

- Multiply each score by its frequency.
Show these products in a **score × frequency** column.
- Find the sum of the **frequency** column.
- Find the sum of the **score × frequency** column.
- **Mean = Sum of the score × frequency column ÷ sum of the frequency column**



Number of toothpicks (Score)	Number of boxes (Frequency)		Score × Frequency
87	2	2 boxes of 87	$2 \times 87 =$
88	8	8 boxes of 88	$8 \times 88 =$
89	14	14 boxes of 89	
90	12	12 boxes of 90	
91	10	10 boxes of 91	
92	3	3 boxes of 92	
93	1	1 box of 93	

1 **a** Complete the **score × frequency** column in the table above.

b What is the sum of the **frequency** column?

c What is the sum of the **score × frequency** column?

d Calculate the mean, to the nearest whole number.

3.5 Calculating the mean

CONTINUED

- 2** The table shows the number of defective switches in each carton in a warehouse.

- a** Complete the *score × frequency* column in the table.

- b** What is the sum of the *frequency* column?

- c** What is the sum of the *score × frequency* column?

Number of defective switches (Score)	Number of cartons (Frequency)	Score × Frequency
0	18	
1	3	
2	5	
3	2	
4	3	
5	1	

- d** Calculate the mean. _____

- 3** Rebecca conducted a survey of the number of children in 40 families. She obtained the following scores.

3	2	1	3	1	0	2	2	1	4
1	2	5	2	0	3	4	2	2	1
2	1	2	3	3	2	1	3	3	2
4	3	3	2	1	0	5	2	3	2



- a** Arrange this data in the following frequency table.

Number of children (Score)	Tally	Frequency	Score × Frequency
0			
1			
2			
3			
4			
5			

- b** Calculate the mean number of children per family. _____

3.6 Using the mean, median and mode

The mean, median and mode are all used to describe the *middle* of a set of scores. This is sometimes called the *central tendency*. Each of these terms can be used to answer a certain type of question.



- The **mean** is the mathematical term for the average of a set of scores. For example, the *average* family in Woodport has 2.4 children.
- The **median** is the **middle score** after the scores are arranged in order from smallest to largest. Half the scores are smaller than the median. The other half are larger than the median. For example, the *middle height* of the students in a class.
- The **mode** is the score that occurs most often. It is the score with the highest frequency. For example, the *most popular television actor* and the *most common dress size*.

1 Which measure – the mean, the median or the mode – would be used in these situations?

- The average speed of cars in a motor race _____
- The most common collar size in shirts _____
- Half of the class were absent for more than 5 days last term _____
- The batting average of a cricketer _____
- The most popular soft drink _____
- The average score in a class test. _____

2 Greg and Richard play golf together each week. The scores for their last ten games are shown below.

Greg:	71	72	73	71	71	73	71	74	72	74
Richard:	72	71	72	71	73	74	73	72	71	72

- What is the mode of Greg's scores? _____
- What is the mean of Greg's scores? _____
- What is the mode of Richard's scores? _____
- What is the mean of Richard's scores? _____
- A prize is awarded to the golfer with the lowest average. Which player collects the prize? _____

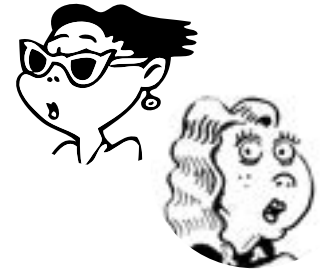
3.6 Using the mean, median and mode

CONTINUED

- 3** Katherine and Michaela play in the same cricket team. Their batting scores so far this season are shown below.

Katherine: 43 39 54 40 56 48 45 48 41

Michaela: 28 0 43 47 19 105 5 49 97



- a** Which player had the highest score in an innings? _____
- b** What is the mean of Katherine's scores? _____
- c** What is the mean of Michaela's scores? _____
- d** Which player has the higher batting average (mean)? _____
- 4** Two students were absent for a class test. The results for the other students were:
12 11 14 13 19 14 12 18 20 12 19 16.
- a** What is the mode of these scores? _____
- b** What is the mean of these scores? _____
- c** What is the median of these scores? _____
- d** Valerie was absent. She normally ranks in the middle of the class. What score would you expect her to receive? _____
- e** Adam was also absent. He is an average student. What score would you expect Adam to receive? _____
- 5** The following table shows the monthly rainfall (in millimetres) recorded in three cities.

City	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Brisbane	167	162	145	88	70	69	57	48	49	75	94	129
Rockhampton	179	167	110	58	41	58	41	22	39	45	71	112
Toowoomba	140	120	99	61	53	61	54	42	48	72	86	118

- a** Calculate the mean monthly rainfall for Brisbane. _____
- b** Calculate the mean monthly rainfall for Rockhampton. _____
- c** Calculate the mean monthly rainfall for Toowoomba. _____
- d** Which city had the highest average monthly rainfall? _____

Teacher's notes

OBJECTIVES

- Draw a picture graph using an appropriate symbol and key.
- Draw a column graph using a suitable scale.
- Draw a line graph to represent data that demonstrates a continuous change (e.g. temperatures at various times during the day).
- Draw a divided bar graph or sector graph to show how a total is divided into parts.

PRIOR KNOWLEDGE

Students should be able to read and interpret information presented in picture graphs, column graphs, divided bar graphs, line graphs and sector graphs. They should be able to mark equal intervals and label the axes of a graph. They should be able to read and interpret data presented in a table.

BACKGROUND

Column graphs may consist of vertical or horizontal columns. Horizontal column graphs are sometimes called bar graphs. This can be confusing for students and is best avoided. The term 'bar graph' is reserved for divided bar graphs in these activity sheets.

STARTER ACTIVITY

Show students a sample of each type of graph covered in this unit. Ask students to name each type of graph and the type of data (numerical or categorical) presented. Then discuss the main features of each graph, such as the heading, the axes, how the axes are labelled and what the scale is. Briefly discuss what each graph shows.

ACTIVITY SHEETS

4.1 Picture graphs

In this activity sheet, students are asked to draw picture graphs to display categorical data. The symbol and key are given for each graph. Students will need to answer question 3 on a sheet of paper or in their workbooks.

Answers

- 1 The half symbol represents \$5.

4.2 Column graphs

In this activity sheet, students are asked to display categorical data in a column graph. The vertical axes are labelled in questions 1 and 2. Students will need a sheet of 5 mm grid paper (provided on page 64) to answer question 3. Only vertical column graphs are featured in this activity sheet.

No answers provided.

4.3 Divided bar graphs

In this activity sheet, students are asked to draw divided bar graphs to display categorical data. Each question is answered on the activity sheet.

No answers provided.

4.4 Line graphs

In this activity sheet, students are asked to draw line graphs to display numerical data. The horizontal and vertical axes are labelled in question 1. A plain grid is provided for question 2. Students will need a sheet of 5 mm grid paper (provided on page 64) to answer question 3.

No answers provided.

Teacher's notes

CONTINUED

4.5 Sector graphs

In this activity sheet, students are asked to calculate the angle for each sector and then draw a sector graph to display categorical data. They will need to answer question 2 on a sheet of paper or in their workbooks.

Answers

1

Vehicles	Fraction	Angle
Trucks	$\frac{1}{3}$	120°
Cars	$\frac{1}{2}$	180°
Motorbikes	$\frac{1}{12}$	30°
Others	$\frac{1}{12}$	30°

2

Type of housing	Fraction	Angle
Houses	$\frac{1}{2}$	180°
Units	$\frac{1}{4}$	90°
Flats	$\frac{1}{6}$	60°
Farms	$\frac{1}{12}$	30°

EXTENSION IDEAS

Students could compare different graphs constructed from the same data and discuss which graph is the most appropriate method of display.

Students could use a spreadsheet program to tabulate and graph data.

4.1 Picture graphs

Nina works at a car sales yard. She drew this picture graph to show the number of cars sold over four weeks.

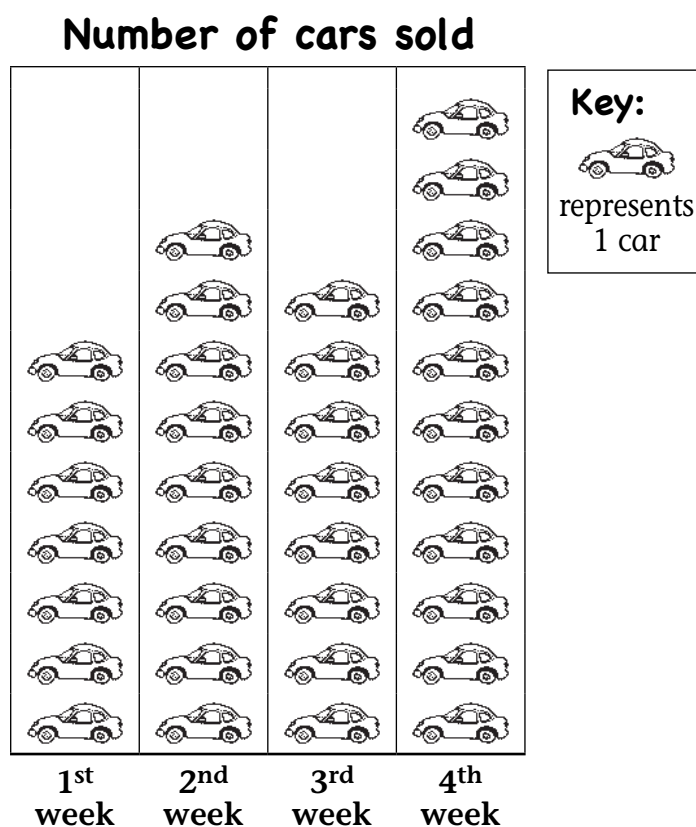
A drawing of a car is used to represent each car sold.

On a picture graph

- Pictures or symbols indicate the number of items in each category.
- The value of each symbol must be shown. This is called the key.
- The symbols should be the same size. They should be equally spaced.
- The graph should have a title or heading.

To draw a picture graph

- Choose a title or heading.
- Choose a picture or symbol.
- State the key. What number does each picture represent?
- Draw the graph. Using grid paper will help keep the symbols in line.



- Kane spent these amounts last week.

Monday: \$25	Thursday: \$20	Saturday: \$40
Tuesday: \$20	Friday: \$35	Sunday: \$30
Wednesday: \$15		

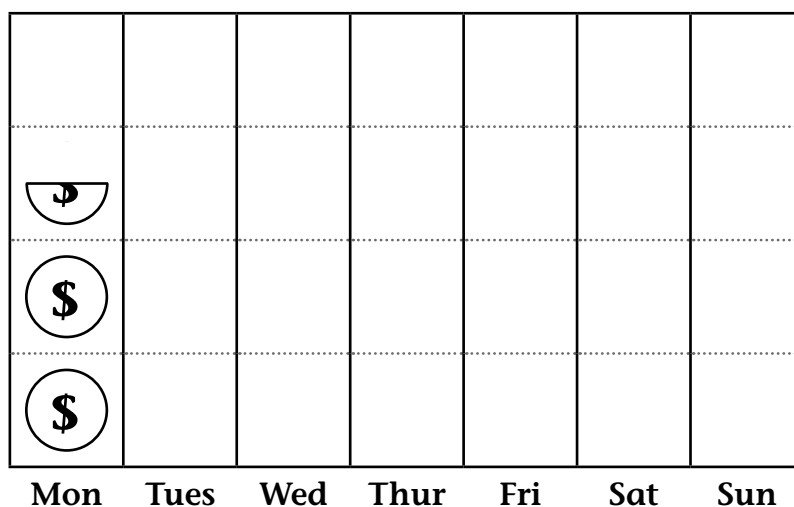
Complete the picture graph to show this data.

Use the symbol  to represent \$10.

The first column has been drawn for you.

What does the symbol  represent? _____

Title → **Kane's weekly spending**



4.1 Picture graphs

CONTINUED

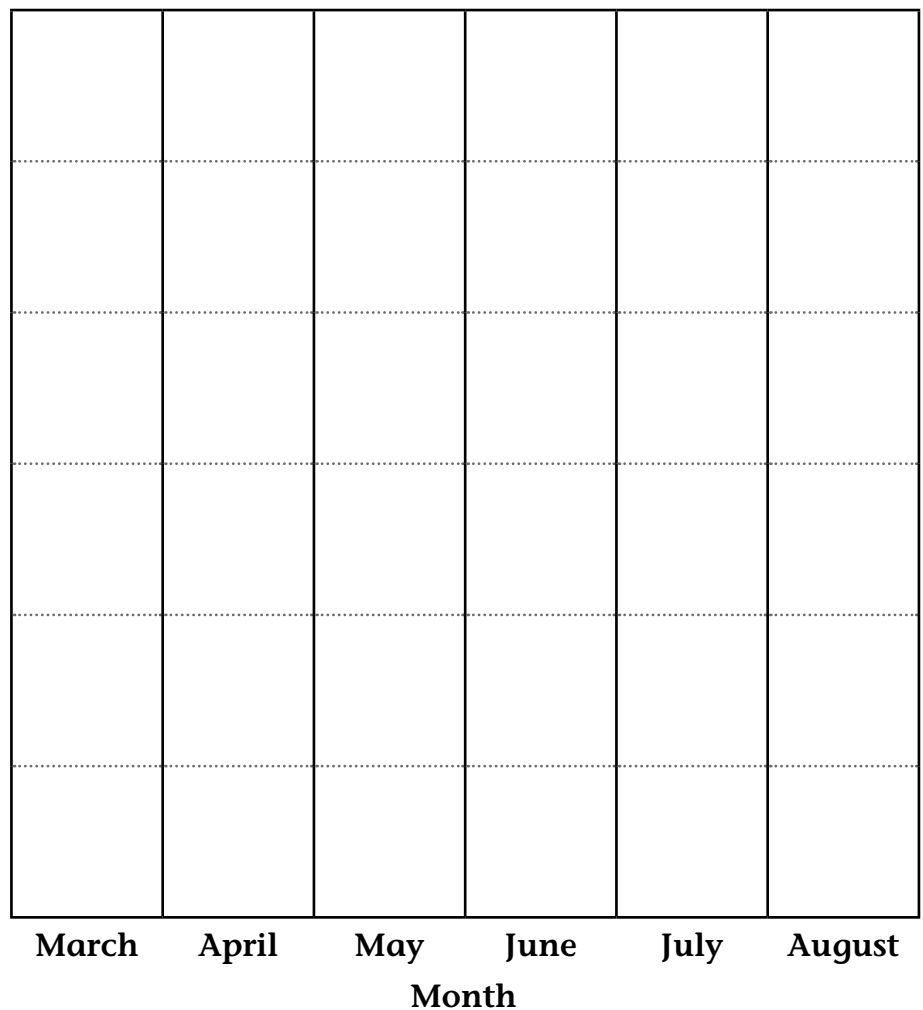
- 2** The table shows the number of new houses completed in Woodport Shire during six months.

Month	Houses completed	Month	Houses completed	Month	Houses completed
March	2	May	6	July	5
April	8	June	9	August	10

Draw a picture graph to show this data.

Title: _____

Use the symbol  to represent 2 houses. Remember to put only 1 symbol in each grid square.



Answer this question
in your workbook.

- 3** The athletics team at Woodport High School is made up of students from each school year. The number of students selected from each year are shown below.

Year 7: 20	Year 8: 30	Year 9: 24
Year 10: 28	Year 11: 22	Year 12: 26

Draw a picture graph to show this data. Use ☺ to represent 4 students.

4.2 Column graphs

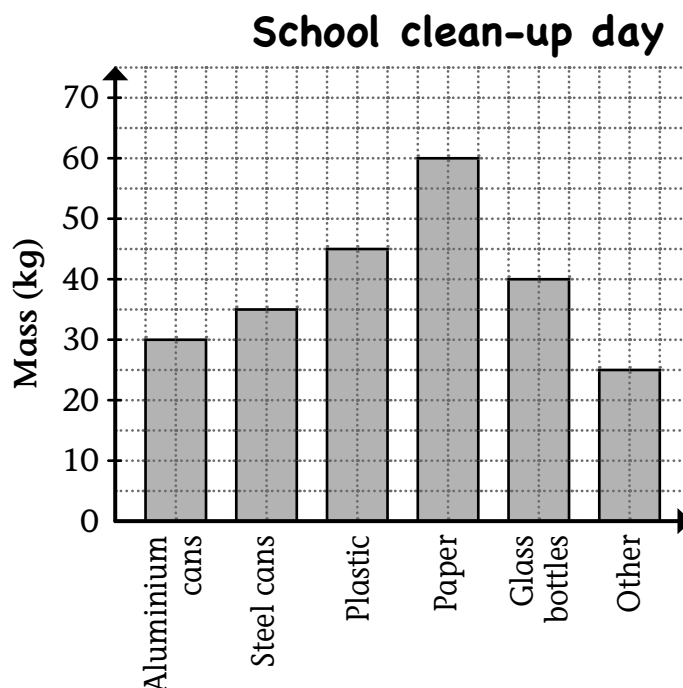
After a school clean-up day, Gillian drew this column graph. It shows the amounts of different types of items that were collected.

On a column graph

- There is a column for each type of item.
- Each column is the same width.
- The columns should be spaced at equal intervals on the horizontal axis.
- You must label both axes.
- A numerical scale on the vertical axis indicates the height of the column.
- The graph should have a title or heading.

To draw a column graph

- Choose a title.
- Choose a scale and label the vertical axis.
- Choose a width for each column. Label the horizontal axis.
- Draw the graph. The space between each column must be the same.



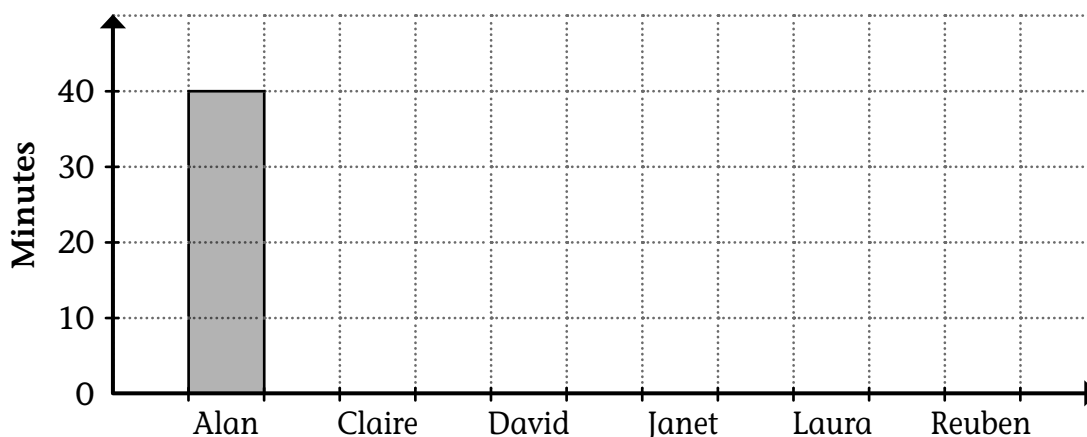
- 1 The times taken for six friends to travel to school each day are shown below.

Alan: 40 minutes Claire: 10 minutes David: 30 minutes
 Janet: 25 minutes Laura: 20 minutes Reuben: 15 minutes

Complete this column graph to show these times.

Use 1 cm to represent 10 minutes. The first column is drawn for you.

Add a title: _____



4.2 Column graphs

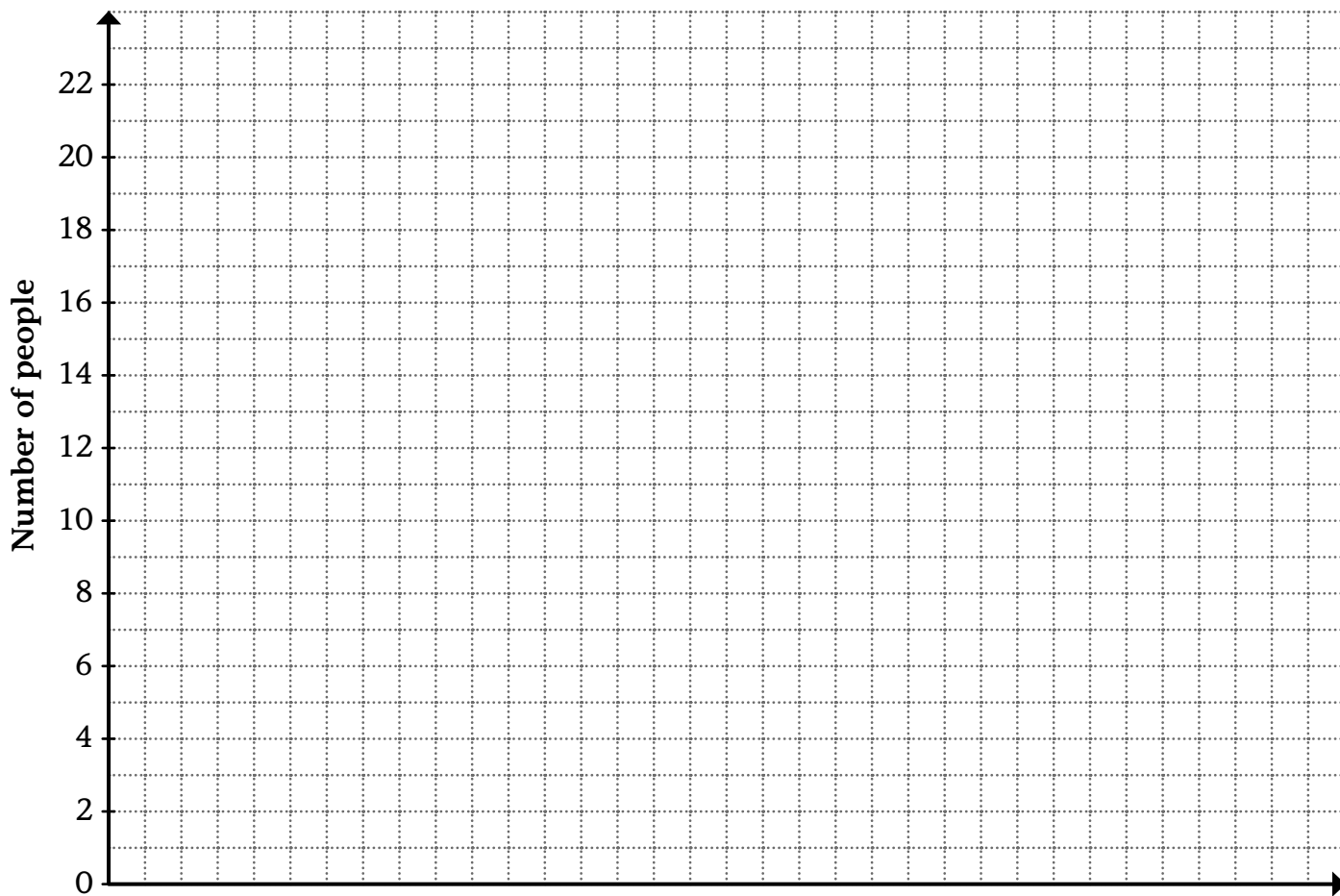
CONTINUED

- 2 Jenny asked 100 people to name their favourite colour. The results are shown in this table.

Blue	Green	Pink	Yellow	Red	Orange	Purple	Others
15	22	20	14	12	6	8	3

Draw a column graph to present this data. A grid is provided for you.

Title: _____



- 3 The daily sales at the local beach store are shown in this table.

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
\$650	\$600	\$750	\$800	\$950	\$1400	\$1350

Draw a column graph to present this data. You will need a sheet of grid paper. You could use 1 cm to represent \$100.

Nadine's pocket money spending

A black and white cartoon illustration of a woman. She has a very large, prominent nose, thin eyebrows, and is wearing glasses. Her mouth is open in a wide, toothy grin. She is wearing a dark-colored top with a row of white polka dots and a pearl necklace. She is holding a large, light-colored shopping bag in front of her. The drawing style is simple and expressive, with thick black lines.

Title: _____

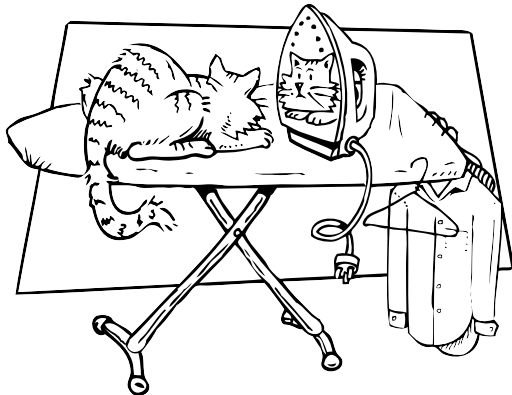
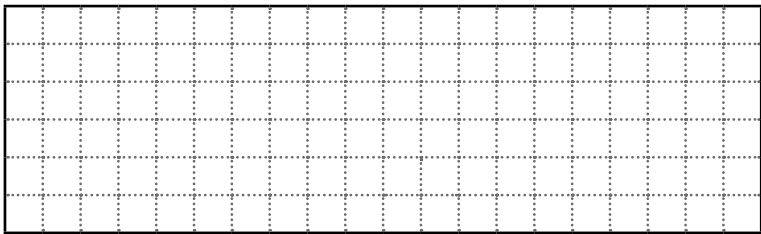


4.3 Divided bar graphs

CONTINUED

- 2** Alison conducted a survey of 100 people. She asked them to name their favourite pet. The results are shown in the table.
- Draw a divided bar graph to present this data on the grid below.
- The grid has 20 units across.
- Let each unit represent 5 people.

Favourite pet	Number of people
Dog	40
Cat	25
Bird	20
Tropical fish	10
Other	5

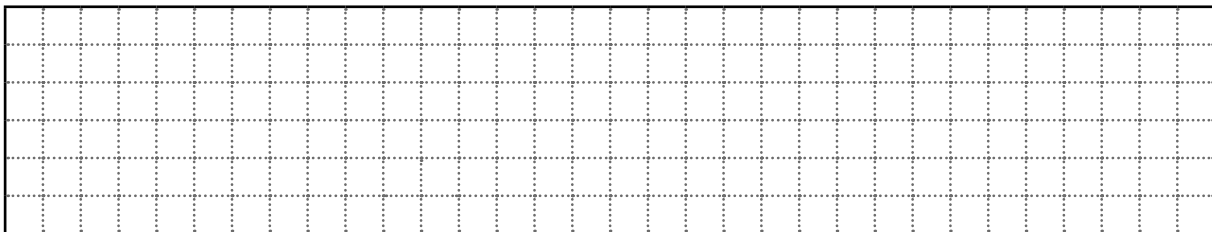


- 3** Students at Woodport High School were given 5 choices for their summer sport: Swimming, Cricket, Softball, Netball and Tennis.
- The choices of 160 students are shown in this table.

Sport	Swimming	Cricket	Softball	Netball	Tennis
Number of students	55	30	30	25	20

Draw a divided bar graph to present this data.

Using 1 mm for each student may make it easier to measure each section.



4.4 Line graphs

Binh rolled two dice 84 times. He recorded the sum of the two numbers on the top faces each roll. The results are shown in this line graph.

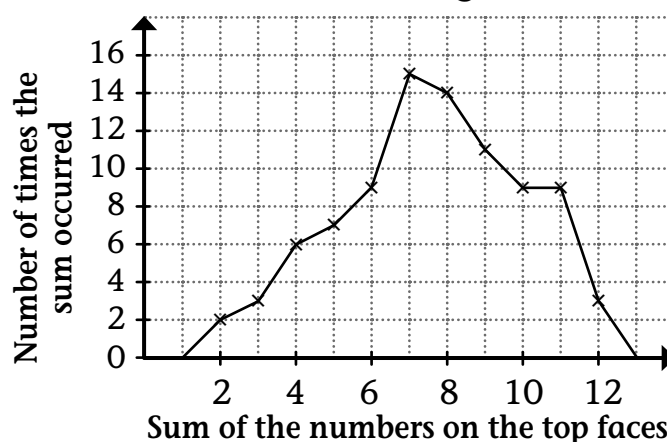
On a line graph

- You must label both axes.
- There is a numerical scale on both axes.
- The graph should have a title or heading.
- A line graph clearly shows a trend or relationship.

To draw a line graph

- Choose a title.
- Choose a scale and name for each axis.
- Plot the points on the grid.
- Join the points with straight lines to complete the graph.

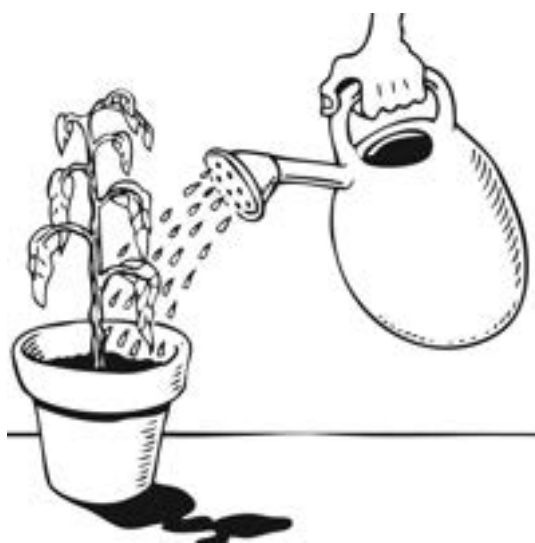
Results of rolling two dice



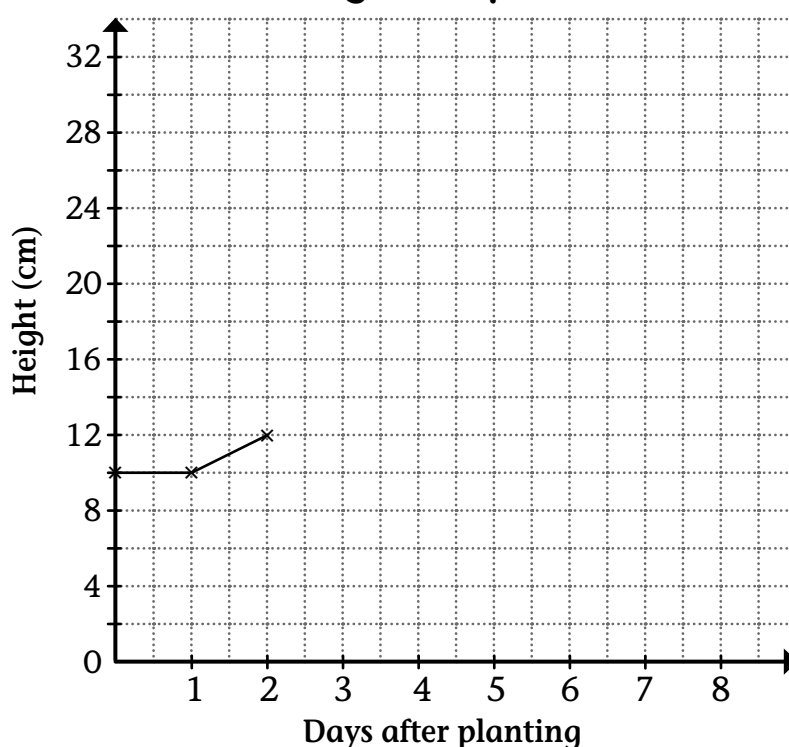
- John recorded the height of a small plant each day. His measurements are shown in this table.

Days after planting	0	1	2	3	4	5	6	7	8
Height (in cm)	10	10	12	15	20	25	28	30	32

Complete this line graph to show the height of the plant. The first 3 points have been plotted for you.



Height of plant



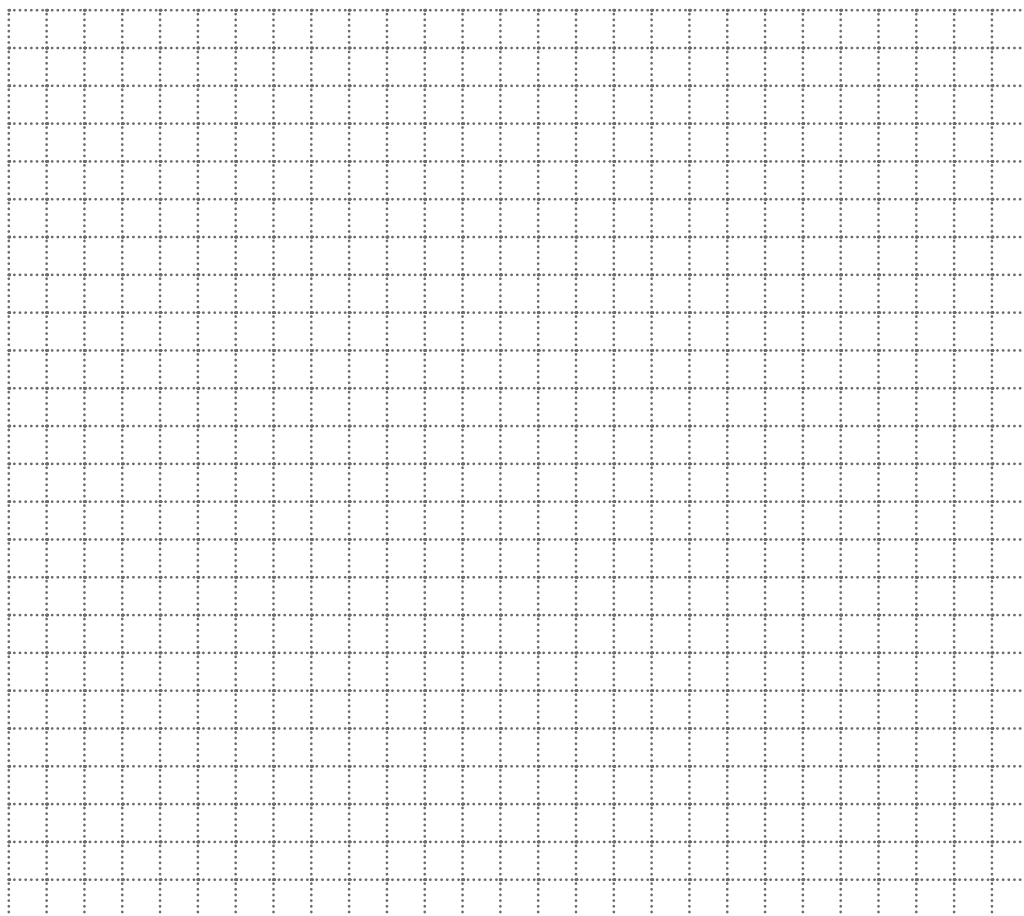
4.4 Line graphs

CONTINUED

- 2** Kayla recorded the temperature at various times during the day. Her records are shown in this table.

Time	8 am	9 am	10 am	11 am	12 pm	1 pm	2 pm	3 pm	4 pm	5 pm
Temperature (°C)	12	14	17	20	22	23	21	19	16	14

Draw a line graph to show this data.



- 3** Joel recorded the maximum temperature each day last week.

Day	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Temperature	30°C	24°C	26°C	29°C	32°C	30°C	28°C

Draw a line graph to present this data. You will need a sheet of grid paper.

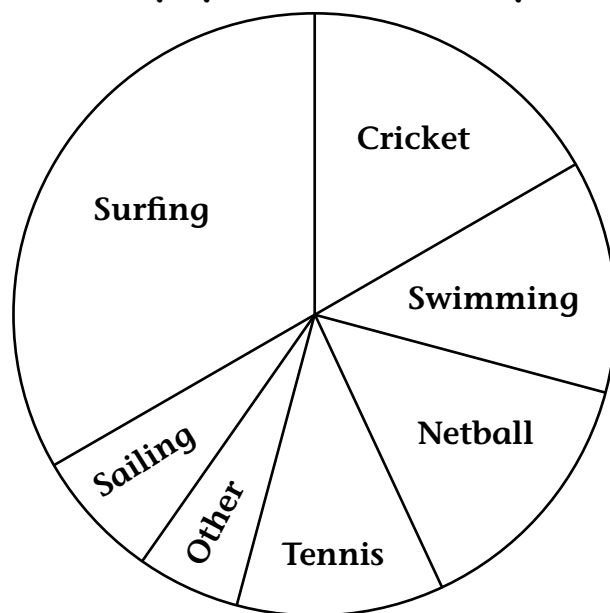
4.5 Sector graphs

Alison conducted a survey to find the most popular summer sport at Woodport High School. She drew this sector graph to show the results of the survey.

A **sector graph** consists of a circle divided into sectors. They are sometimes called *pie graphs* because the sectors look like pie slices.

Sector graphs show how a whole thing can be divided into parts. You need to use a **protractor** to measure the angles in each sector.

Most popular summer sport



To draw a sector graph

- Choose a title.
- Calculate the size of the angle in each sector.
- Choose a suitable radius for the circle.
- Draw the graph. Each sector must be clearly labelled.

- 1 Lisa and Tim conducted a survey of the vehicles that passed their school in one hour.

They counted 600 vehicles in one hour.

Their results are shown in this table.

Vehicles	Number	Fraction	Angle
Trucks	200	$\frac{200}{600} = \frac{1}{3}$	$\frac{1}{3} \times 360 = 120^\circ$
Cars	300		
Motorbikes	50		
Others	50		

To draw a sector graph you need to calculate the size of the angle in each sector.

For example, trucks represented $\frac{200}{600} = \frac{1}{3}$ of the vehicles counted.

A sector that is $\frac{1}{3}$ of a circle contains an angle measuring $\frac{1}{3} \times 360^\circ = 120^\circ$.

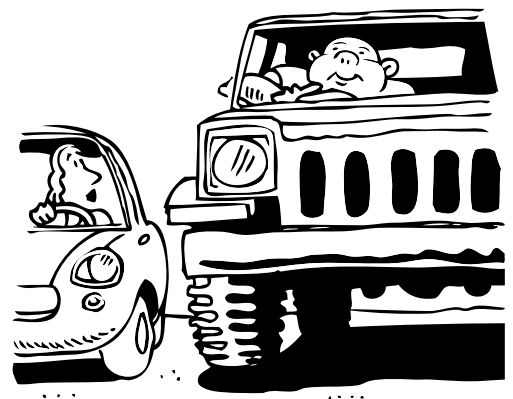
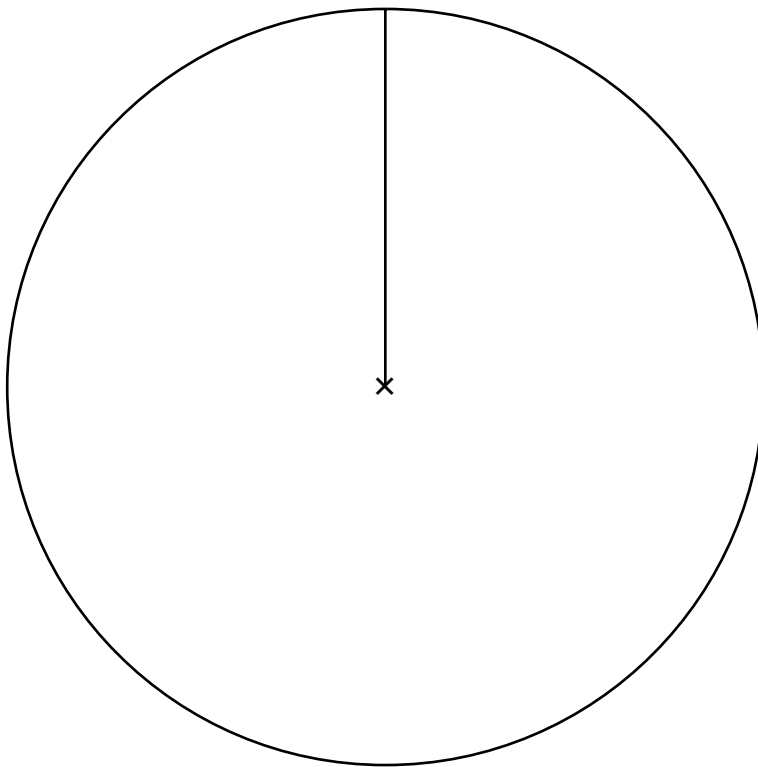
- a Calculate the angles in the other sectors and complete the table.

4.5 Sector graphs

CONTINUED

- b Complete this sector graph to show the results of the survey.
Use a **protractor** to measure the angle in each sector.

Title: _____



- 2 The town of Woodport has an adult population of 3 600 people.
The type of housing they live in is shown in the table.

Type of housing	Number of people	Fraction	Angle
Houses	1 800		
Units	900		
Flats	600		
Farms	300		

On a sheet of paper, construct a circle with radius 5 cm.

Mark the centre of the circle.

Complete a sector graph to show this data.

Use the data in the table to calculate the angle in each sector.

Teacher's notes

OBJECTIVES

- Read and interpret data presented in a dot plot.
- Draw a dot plot to show a small set of numerical data.
- Read and interpret frequency histograms and polygons.
- Draw frequency histograms and polygons to present numerical data.

PRIOR KNOWLEDGE

Students should be able to read and interpret data presented in a table. They should be able to mark equal intervals and label the axes of a graph.

BACKGROUND

Students should be encouraged to read carefully the question that is asked. They should only add a frequency polygon to a histogram when asked to do so. If a question only asks them to draw a frequency polygon, then a histogram should not be drawn. The two graphs should not automatically be drawn together.

STARTER ACTIVITY

Show students a sample of each type of graph covered in this unit. Ask students to name each type of graph and the type of data (numerical or categorical) presented. Then discuss the main features of each graph, such as the heading, the axes, how the axes are labelled and what the scale is? Briefly discuss what each graph shows.

ACTIVITY SHEETS

5.1 Dot plots

In this activity sheet, students read and interpret data presented in simple dot plots. On the second page, they are asked to draw a dot plot to present a small set of numerical data.

Answers

- 1 a 6 b 4 c 1 d 4 e 3
 2 a 7 b 3 c 2 d 4 e 4
 3 a 4 b 0
 4 a 8 b 1

5.2 Frequency histograms

In this activity sheet, students read and interpret data presented in a frequency histogram. They are also asked to arrange the data from the graph in a frequency table.

Answers

- 1 3 2 5 3 8 4 0
 5

Days absent	Number of students
0	8
1	3
2	5
3	9
4	0
5	5

- 6 4 7 11 8 3 9 52

10

Age	Frequency
12	4
13	6
14	12
15	11
16	9
17	5
18	3
19	2

Teacher's notes

CONTINUED

5.3 Drawing frequency histograms

In this activity sheet, students are asked to draw a frequency histogram to present a set of numerical data given in a frequency table. The horizontal and vertical axes are labelled in question 1. A plain grid is provided for question 2. Students will need a sheet of 5 mm grid paper (provided on page 64) to answer question 3.

No answers provided.

5.4 Frequency polygons

In this activity sheet, students read and interpret data presented in a frequency polygon. They are also asked to arrange the data from the graph in a frequency table.

Answers

1 9 2 4 3 8 4 36

5

Score	Frequency
0	8
1	6
2	9
3	4
4	5
5	3
6	1

6 7 7 2 8 3 9 7 10 42

11

Favourite number	Frequency
1	3
2	2
3	4
4	5
5	4
6	6
7	8
8	7
9	3

5.5 Drawing frequency polygons

In this activity sheet, students are asked to draw a frequency polygon to present a set of numerical data given in a frequency table. The horizontal and vertical axes are labelled in question 1.

A plain grid is provided for question 2. Students will need a sheet of 5 mm grid paper (provided on page 64) to answer question 3. In question 3, they are first asked to draw a frequency histogram then add a frequency polygon to the graph.

No answers provided.

EXTENSION IDEAS

Students could discuss and interpret graphs found in the media and various texts.

They could use a spreadsheet program to tabulate and graph data.

5.1 Dot plots

Sue is the goal shooter in her netball team. This dot plot shows the number of goals she scored in the last 14 matches.

A **dot plot** is like a column graph.

It is used for small sets of **numerical** data.

The scores are equally spaced along a horizontal number line.

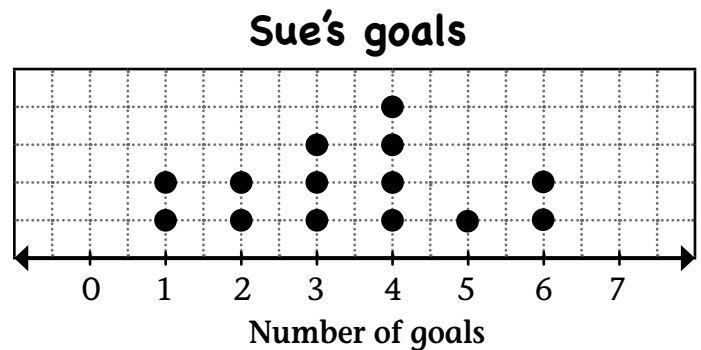
The dots should be equally spaced vertically.

Each dot represents one result.

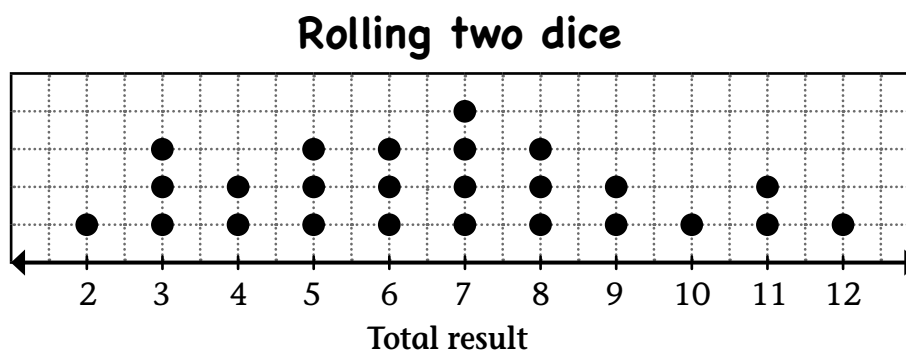
So, Sue scored 6 goals in 2 matches.

A dot plot makes it easy to see where the scores are clustered together.

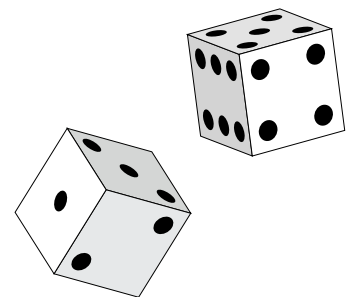
It is also easy to see gaps in the data.



- 1 Use the dot plot above to answer these questions.
 - a What was the highest number of goals Sue scored in a match? _____
 - b In how many matches did Sue score 4 goals? _____
 - c In how many matches did Sue score 5 goals? _____
 - d In how many matches did Sue score less than 3 goals? _____
 - e In how many matches did Sue score more than 4 goals? _____
- 2 Richard rolled two dice 25 times. He recorded the total result after each throw. This dot plot shows the recorded results.



- a Which total occurred most? _____
- b How many times was the total result 5? _____
- c How many times was the total result 11? _____
- d How many times was the total result less than 4? _____
- e How many times was the total result more than 9? _____



5.1 Dot plots

CONTINUED

- 3** The Woodport Wolves football team scored the following numbers of goals per game last season:

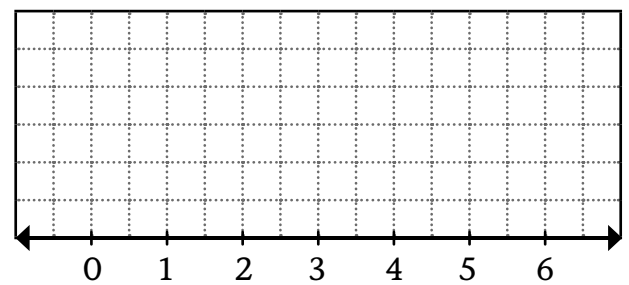
1, 0, 1, 2, 1, 0, 2, 3, 1, 0, 4, 2, 1, 2, 3.

a What was the highest number of goals the team scored in a game? _____

b What was the lowest number of goals the team scored in a game? _____

c Display this data in a dot plot.

Using a grid helps to keep the dots equally spaced vertically.



- 4** A fishing club held a competition last weekend.

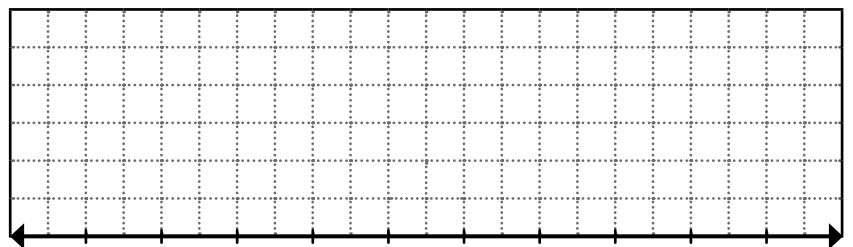
The numbers of fish caught by the competitors were:

4, 2, 5, 6, 1, 3, 4, 5, 2, 7, 1, 4, 5, 6, 3, 8, 2, 4.

a What was the highest number of fish caught? _____

b What was the lowest number of fish caught? _____

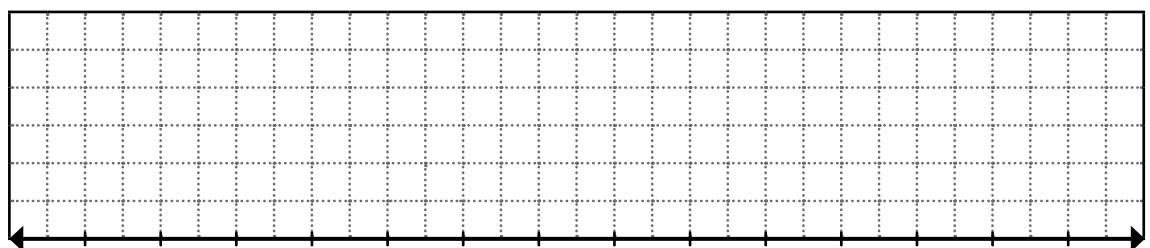
c Display this data in a dot plot.



- 5** Last Saturday's newspaper contained 16 advertisements for used car businesses. The number of cars listed for sale in each advertisement was:

4, 7, 10, 3, 7, 6, 12, 14, 8, 6, 14, 8, 6, 5, 9, 12.

Display this data in a dot plot.



5.2 Frequency histograms

Ms Nguyen drew this graph to show the number of days students were absent last term. This type of graph is called a **frequency histogram**.

Frequency histograms are similar to vertical column graphs. They are used with **numerical** data. They show how many times a result has happened.

The scores (results) are equally spaced along the horizontal axis.

The vertical axis shows the frequency of the scores.

The **frequency** of each score is shown by the length of the rectangles.

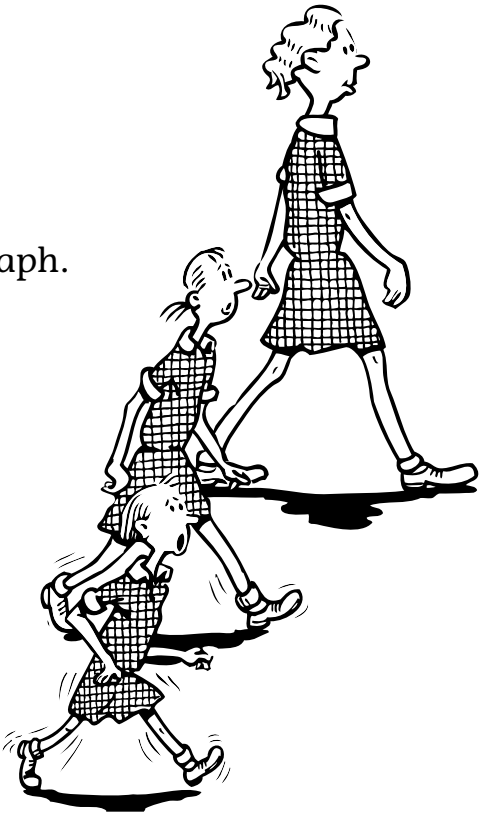
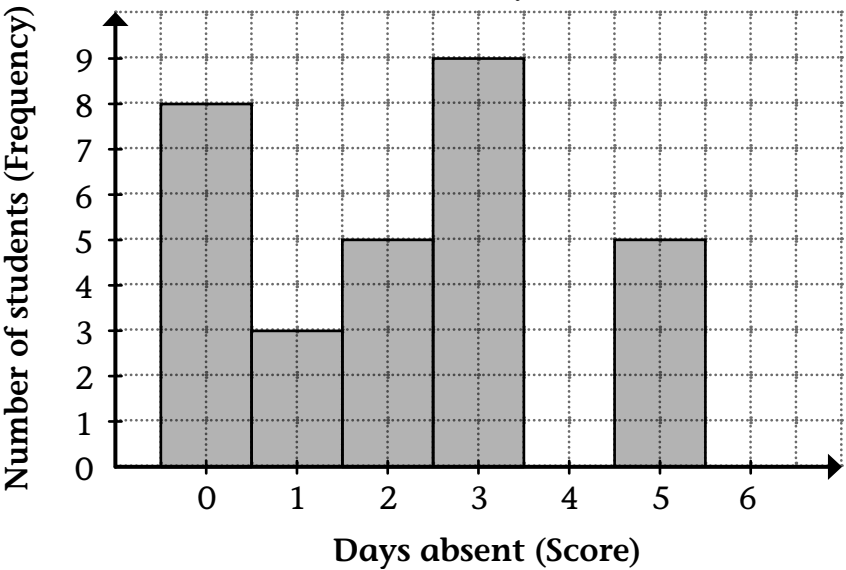
The rectangles are the same width. There are no spaces between the columns.

Use the graph to answer these questions.

- 1 How many students had 1 day absent? _____
- 2 How many students had 2 days absent? _____
- 3 How many students had no absences? _____
- 4 How many students had 4 days absent? _____
- 5 Complete this table to show the data from the graph.

Days absent (Score)	Number of students (Frequency)
0	
1	
2	
3	
4	
5	

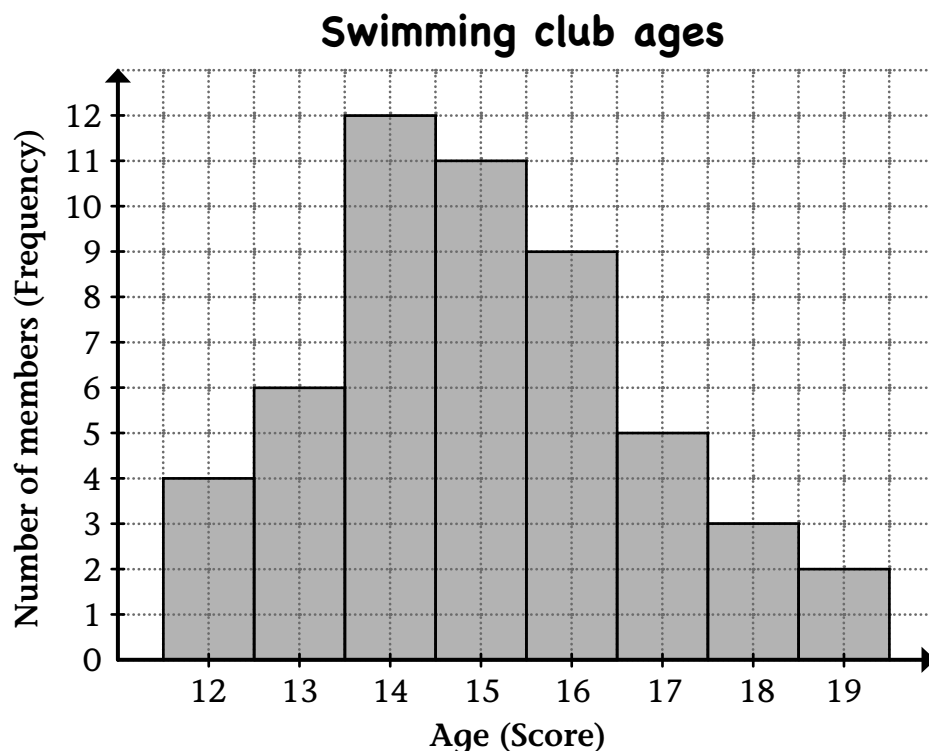
Students' days absent



5.2 Frequency histograms

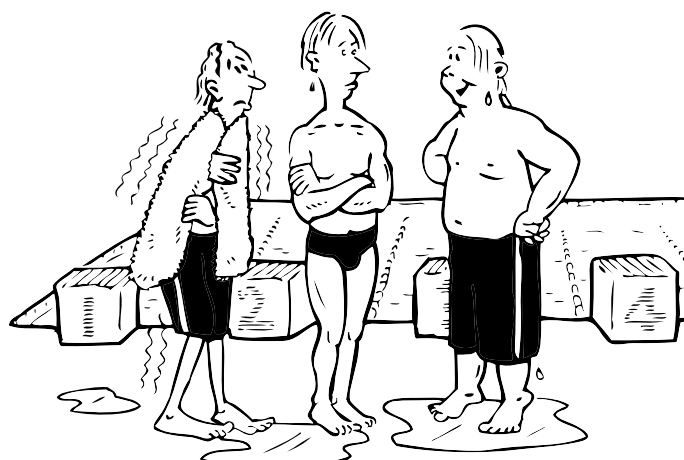
CONTINUED

Adrienne drew this frequency histogram to show the ages of the members of her swimming club.



- 6 How many members of the club are 12 years old? _____
- 7 How many members are 15 years old? _____
- 8 How many members are 18 years old? _____
- 9 How many people are members of the swimming club? _____
- 10 Complete this table to show the data from the graph.

Age (Score)	Number of members (Frequency)
12	
13	
14	
15	
16	
17	
18	
19	



5.3 Drawing frequency histograms

Brett’s class conducted an experiment. Each student tossed a coin 6 times. Brett recorded the number of ‘heads’ in this table.

Number of heads (Score)	0	1	2	3	4	5	6
Frequency	2	3	5	7	6	4	1

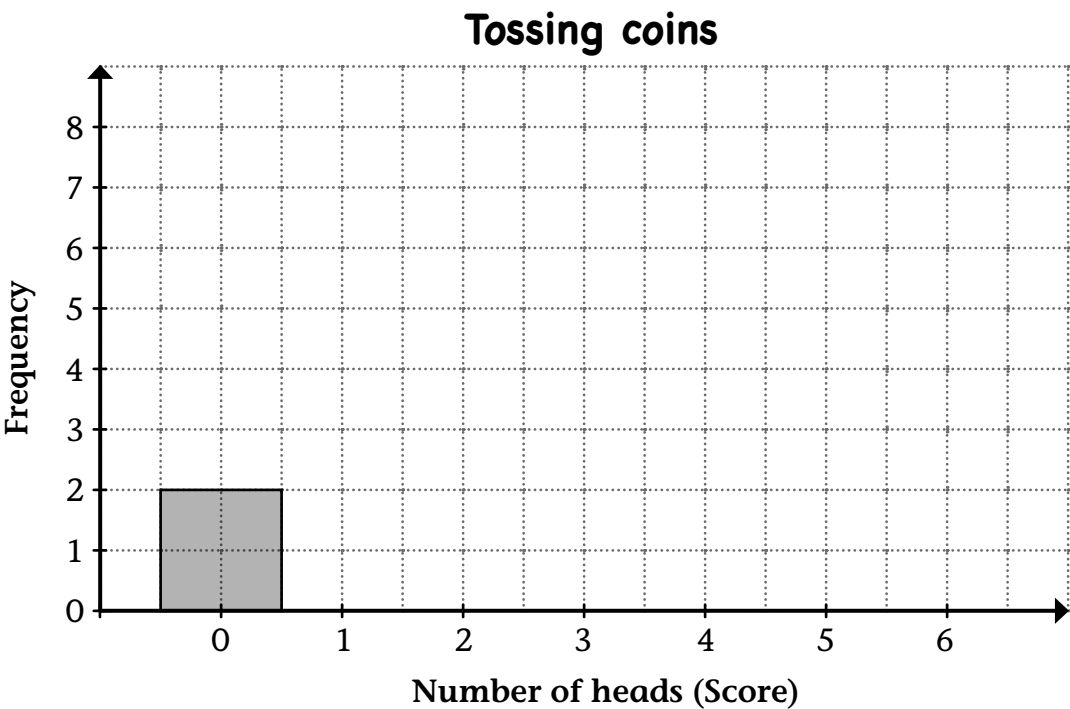
Brett wanted to draw a frequency histogram to show this data.

To draw a frequency histogram:

- Choose a title for the graph.
- Choose a width for each column.
Label the horizontal axis.
- Choose a scale and label the vertical axis.
- The columns are centred on each score.
- The first column begins half a column width from the vertical axis.
- Do **not** leave a gap between the columns.



- 1 Complete this frequency histogram to show Brett’s data.
The first column has been drawn for you.



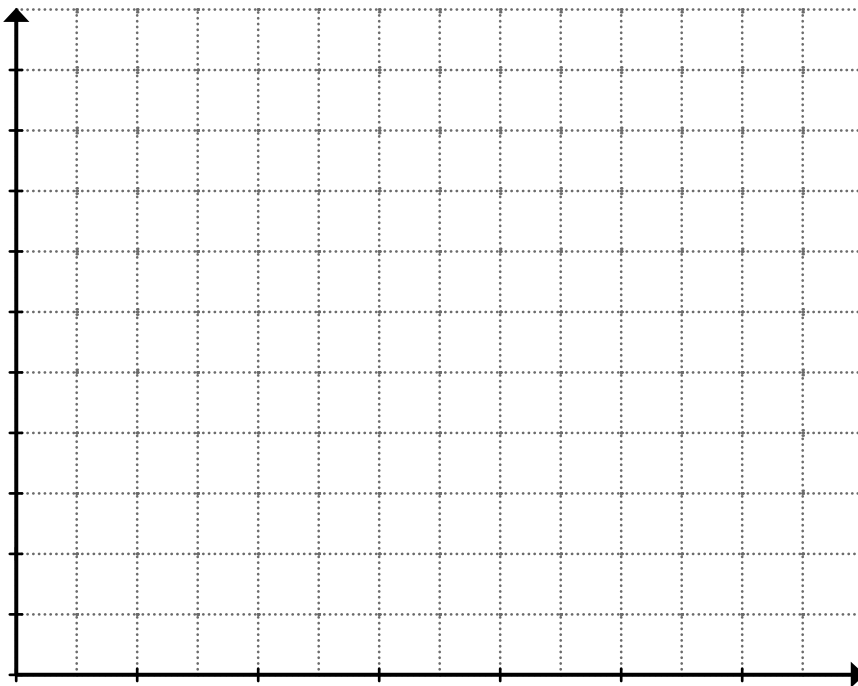
5.3 Drawing frequency histograms

CONTINUED

- 2** Kurt played 15 matches of cricket last season. He bowled in 27 innings. The number of wickets Kurt took in an innings is recorded in this table.

Draw a frequency histogram to show this data.

Wickets (Score)	Innings (Frequency)
1	3
2	5
3	8
4	6
5	4
6	1



You need 5 mm grid paper to complete question 3.

- 3** Carla conducted a survey of 42 families. She recorded the number of children in each family. The results are shown in this table.

Draw a frequency histogram to show this data.

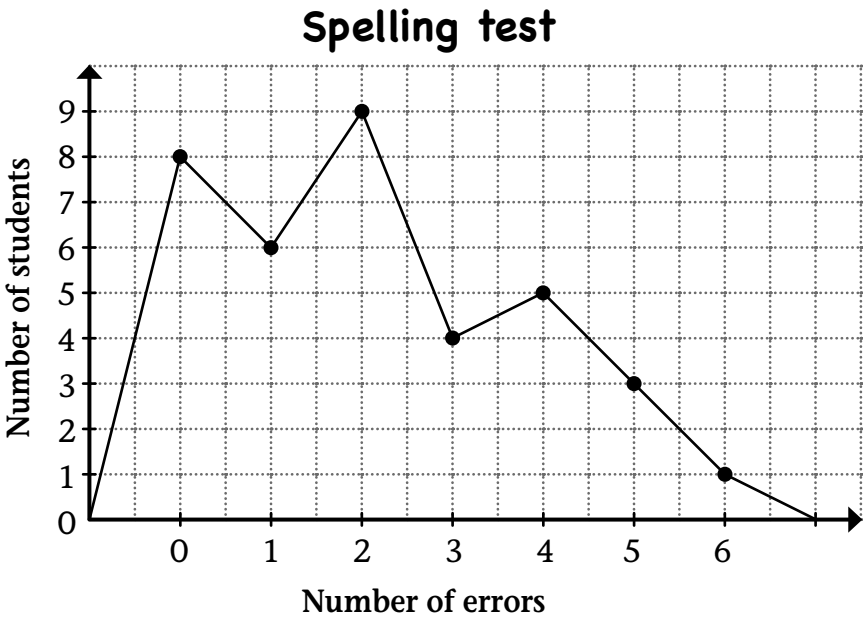
Number of children (Score)	Number of families (Frequency)
0	4
1	12
2	14
3	8
4	3
5	1

5.4 Frequency polygons

Mr Euclid drew this graph to show the number of errors a group of students made in a spelling test.

This type of graph is called a **frequency polygon**.

Frequency polygons are a type of line graph. The points showing the data are joined by straight lines. The first and last points are connected to the horizontal axis.

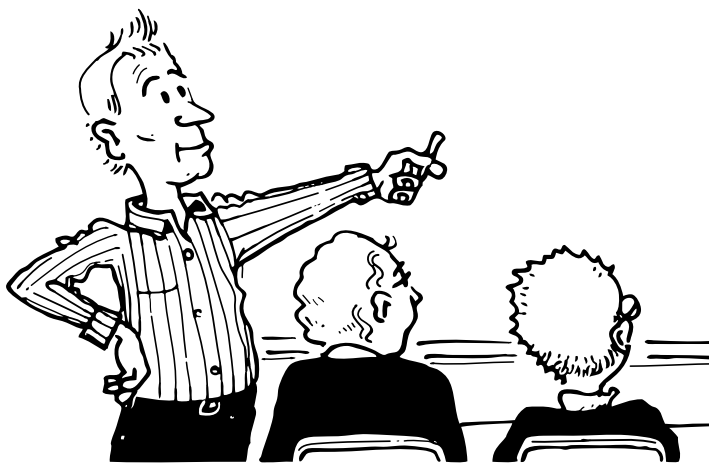


Frequency polygons are used with **numerical** data. They show how many times a result has happened. The scores (results) are equally spaced along the horizontal axis. The vertical axis shows the frequency of the scores.

Use the graph to answer these questions.

- 1 How many students made 2 errors? _____
- 2 How many students made 3 errors? _____
- 3 How many students made no errors? _____
- 4 How many students did the test? _____
- 5 Complete this table to show the data from the graph.

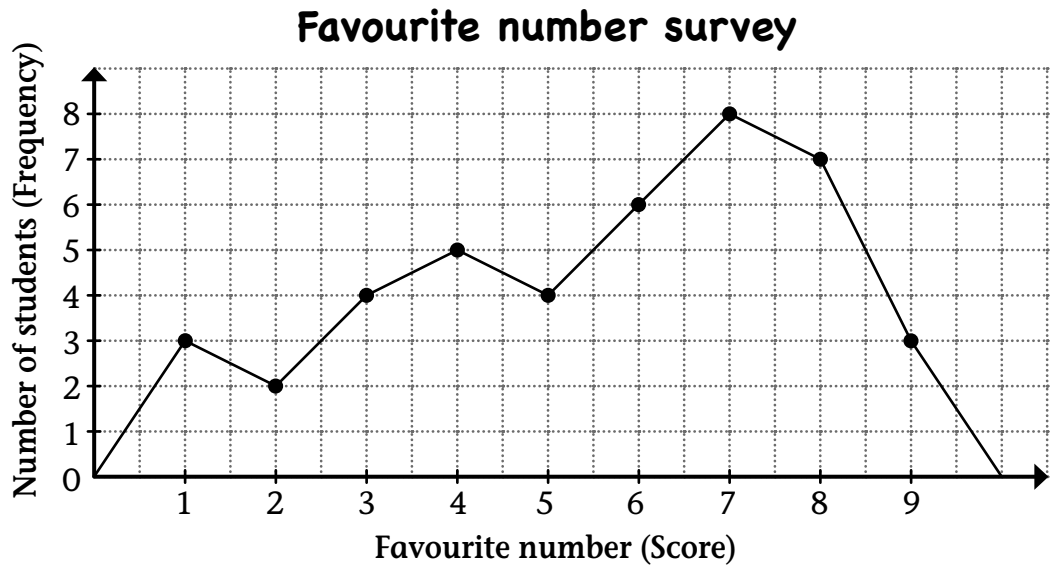
Number of errors (Score)	Number of students (Frequency)
0	
1	
2	
3	
4	
5	
6	



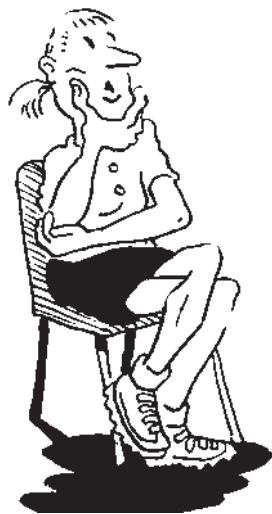
5.4 Frequency polygons

CONTINUED

Melissa asked some students to name their favourite number. She drew this frequency polygon to show the results.



- 6 Which number was most popular? _____
- 7 Which number was least popular? _____
- 8 How many students named number 1? _____
- 9 How many students named number 8? _____
- 10 How many students did Melissa ask to name their favourite number? _____
- 11 Complete this table to show the data from the graph.



Favourite number (Score)	Number of students (Frequency)
1	
2	
3	
4	
5	
6	
7	
8	
9	

5.5 Drawing frequency polygons

This table shows the number of faulty items found in boxes of 100 light globes.

Number of faulty globes (Score)	0	1	2	3	4	5	6	7	8	9	10
Number of boxes(Frequency)	14	6	4	5	2	3	1	2	0	2	1

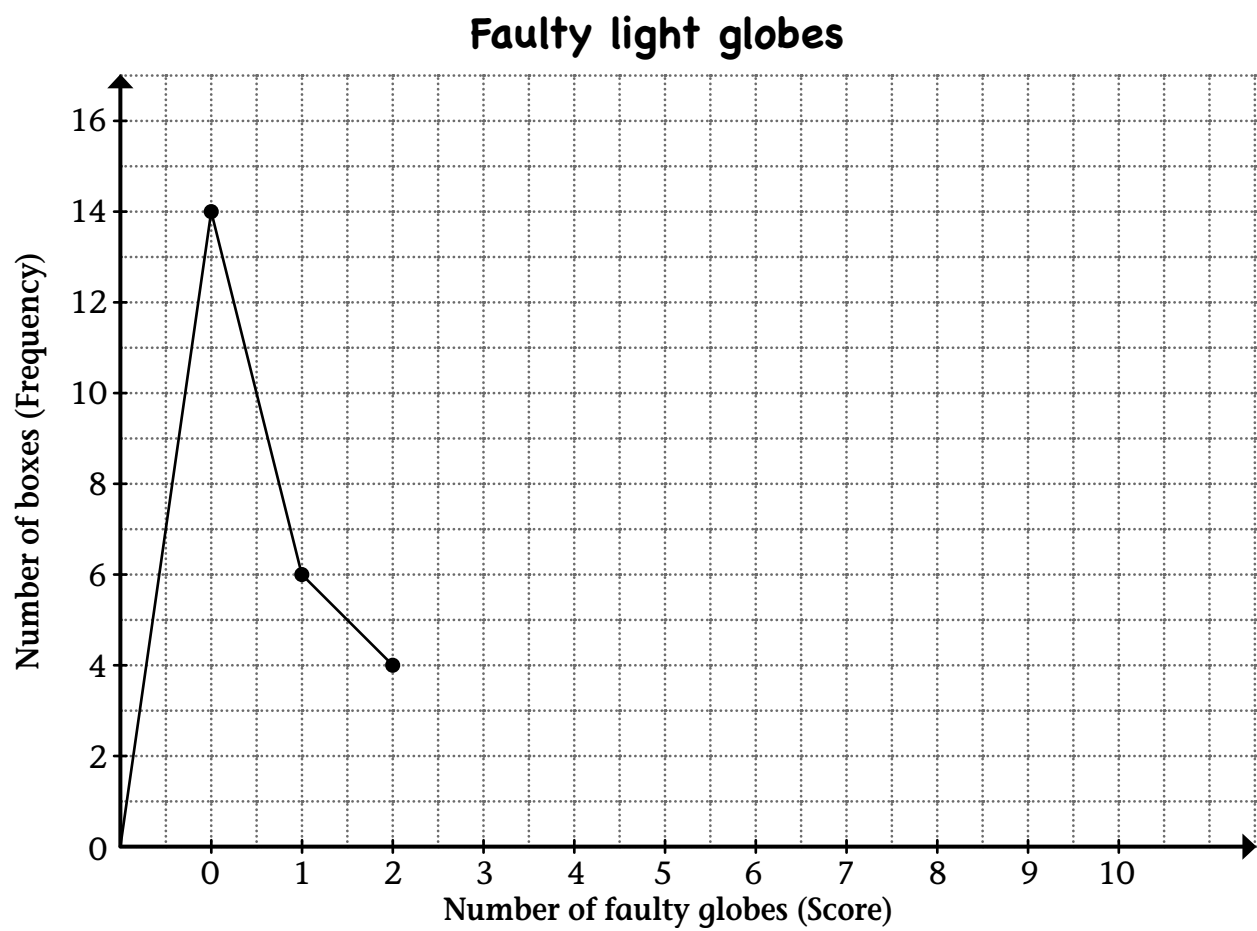
Jaeden wanted to draw a frequency polygon to show this data.

To draw a frequency polygon

- Choose a title for the graph.
- Choose a scale and label the horizontal axis.
- Choose a scale and label the vertical axis.
- The first score is one unit to the right of the vertical axis.
- Plot the points to represent each ordered pair in the table.
- Join these points using straight lines.
- Connect the first and last points to the horizontal axis.



- 1 Complete this line graph to show the data in the table.
The first 3 points have been plotted for you.



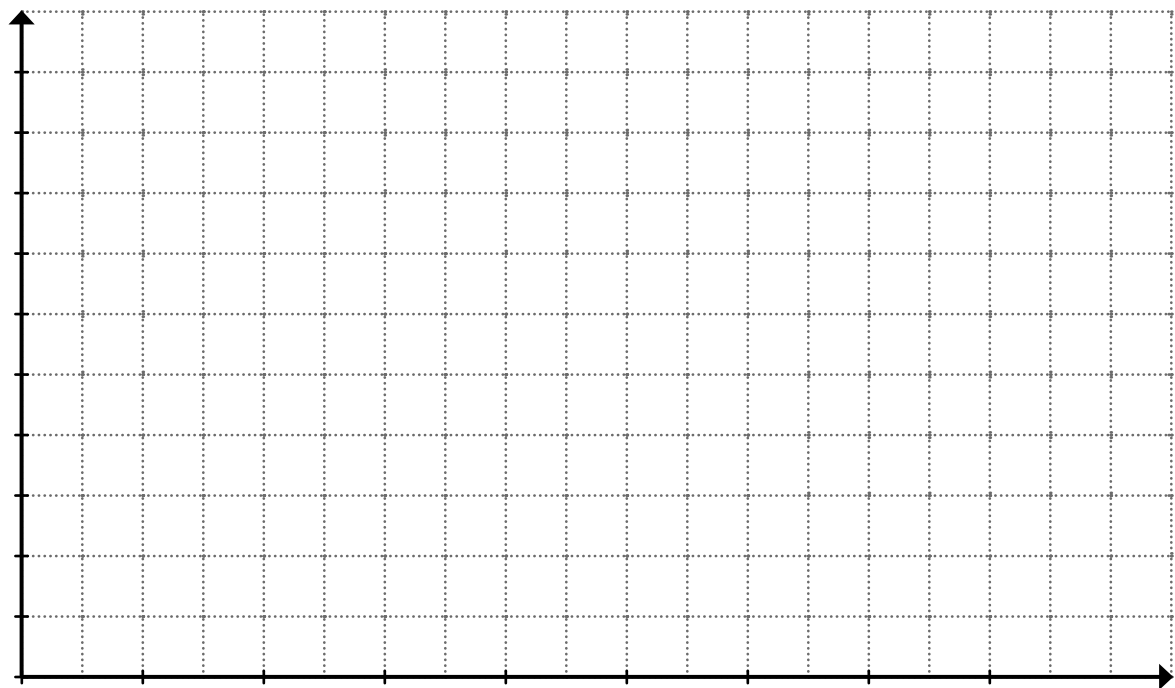
5.5 Drawing frequency polygons

CONTINUED

- 2 Catherine conducted a survey of the number of people living in each house in Wattle Drive. The results are shown in this table.

Number of people (Score)	1	2	3	4	5	6	7	8
Number of houses (Frequency)	3	5	8	10	7	2	0	1

Draw a frequency polygon to present this data.



You need 5 mm grid paper to complete question 3.

- 3 Adam recorded the masses of 35 students in his year at school. The masses were rounded to the nearest 2 kg. He presented the data in this table.

Mass (kg) (Score)	40	42	44	46	48	50	52	54	56	58	60
Number of students (Frequency)	2	1	3	4	6	5	6	4	2	1	1

- a Draw a **frequency histogram** to present this data.

- b Then add a **frequency polygon** to your graph.

To do this, first mark the midpoint of the top of each rectangle. Then join these points with straight lines to form the polygon. Connect the first and last points to the horizontal axis.

Teacher's notes

OBJECTIVES

- Conduct a survey to collect data.
- Organise the collected data in a frequency table.
- Analyse the data using the mean and mode.
- Draw a graph to display the data.

PRIOR KNOWLEDGE

Students should be familiar with frequency tables and the various types of graphs used in these activity sheets. They should be able to identify the mode and calculate the mean for a set of data.

BACKGROUND

This unit allows students to organise and analyse data they have collected. The activity sheets may be used as a unit of work to revise and reinforce the students' knowledge and skills. Alternatively, they may be used independently, using data collected from the class at the beginning of the lesson.

STARTER ACTIVITY

The teacher could discuss the survey questions with the class. They could ask questions like: *What type of data (numerical or categorical) will be collected? Which graphs could be used to display the data collected?*

ACTIVITY SHEETS

6.1 Collecting your data

This activity sheet provides a survey sheet for collecting data from the students' own class or a larger group. The data collected is presented and analysed in the following activity sheets.

There is one activity sheet for each question in the survey.

6.2 Birthdays

In this activity sheet, students analyse the responses to the survey question: *On which day of the week is your birthday this year?* They are asked to calculate the percentage of students who have a birthday falling on each day of the week and draw a divided bar graph to display the data.

6.3 Lucky numbers

In this activity sheet, students analyse the responses to the survey question: *What is your favourite number, or lucky number, less than 10?* They are asked to present the data in a frequency histogram. The teacher could provide students with a sheet of 5 mm grid paper (provided on page 64) to use for their graph.

6.4 Family pets

In this activity sheet, students analyse the responses to the survey question: *Which pets do you have at home?* They are asked to present the data in a column graph. The teacher could provide students with a sheet of 5 mm grid paper (provided on page 64) to use for their graph.

6.5 People at home

In this activity sheet, students analyse the responses to the survey question: *How many people live in your home?* They are asked to present the data in a frequency histogram. The teacher could provide students with a sheet of 5 mm grid paper (provided on page 64) to use for their graph.

6.6 Favourite sport

In this activity sheet, students analyse the responses to the survey question: *What is your favourite sport?* They will need to list the most likely responses in the first column of the frequency table. (This could form part of a class discussion before the survey responses are organised.) They are asked to present the data in a sector graph on the activity sheet.

If the data has been collected from a group containing both girls and boys, they could organise and analyse the data for boys and girls separately. They could then discuss the differences and similarities. Each student will need two copies of the activity sheet for this analysis.

6.7 Household chores

In this activity sheet, students organise and then analyse the responses to the survey question: *How many hours did you spend doing household chores last week?* They are asked to present the data in a frequency histogram and then add a frequency polygon to the graph. The teacher could provide students with a sheet of 5 mm grid paper (provided on page 64) to use for their graph.

Data for boys and girls could be analysed separately, as described for Activity sheet 6.6.

EXTENSION IDEAS

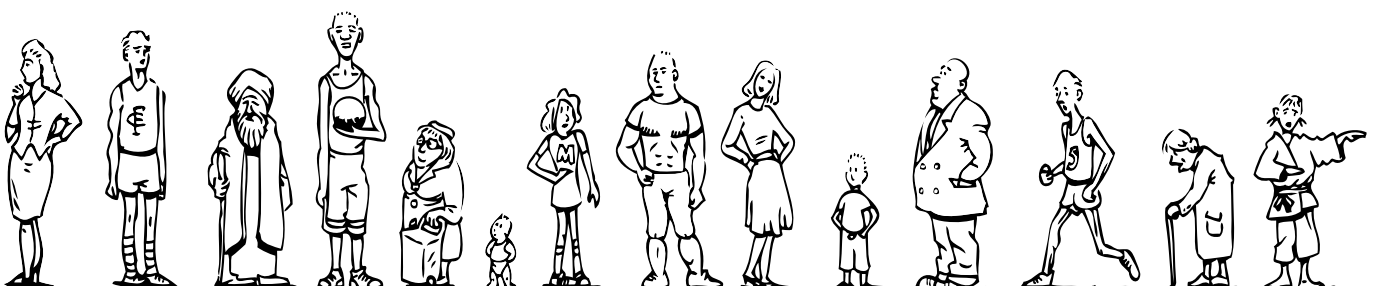
This unit provides an opportunity for students to collect and analyse information about a variety of areas of interest. They could compose their own questions to collect data in the survey.

Tools such as databases, spreadsheets and other software packages may be used where appropriate to organise, display and analyse data.

6.1 Collecting your data

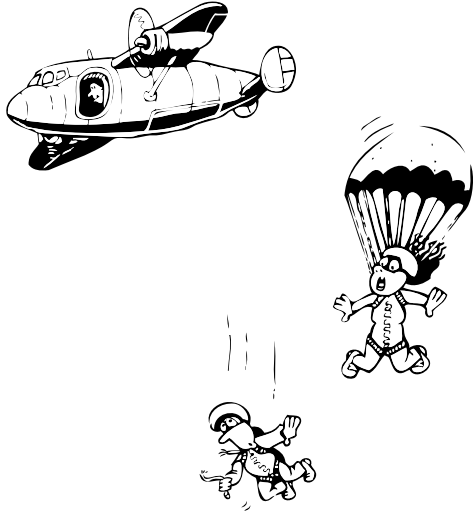
Ask each student to complete one of these survey forms.

CLASS SURVEY <i>Please answer each question in the right hand column.</i>	Tick one: ☐ Male ☐ Female
On which day of the week is your birthday this year?	Monday
	Tuesday
	Wednesday
	Thursday
	Friday
	Saturday
	Sunday
What is your favourite number, or lucky number, less than 10?	
Which pets do you have at home? You may tick more than one answer to this question.	Dog
	Cat
	Bird
	Fish
	Other
	None
How many people live in your home?	
What is your favourite sport?	
How many hours did you spend on household chores last week?	



6.2 Birthdays

- 1 Organise the data from your survey in this table.



Day of birthday

Day	Tally	Frequency
Monday		
Tuesday		
Wednesday		
Thursday		
Friday		
Saturday		
Sunday		

- 2 a How many students were counted in this survey? _____
- b How many students have a birthday on Monday? _____
- c Which day is the mode (most common)? _____
- d How many students have a birthday on Saturday or Sunday? _____
- e How many students have a birthday on a weekday (Monday to Friday)? _____

- 3 Calculate the percentage of the students who have a birthday on each day. (Divide the frequency by the total number of students. Then multiply by 100.)

Day	Frequency	Percentage
Monday		
Tuesday		
Wednesday		
Thursday		
Friday		
Saturday		
Sunday		

- 4 Draw a **divided bar graph** to show this data.

To draw a divided bar graph

- Choose a title.
- Choose a suitable length for the graph. Using 1 mm for each percentage unit can make it easy to measure each section. Your graph will be 100 mm (or 10 cm) long.
- Draw the graph. Each section must be clearly labelled.

6.3 Lucky numbers

- 1 Organise the data from your survey in this table.



Favourite or lucky numbers

Number	Tally	Frequency
1		
2		
3		
4		
5		
6		
7		
8		
9		

- 2
- How many students were counted in this survey? _____
 - How many students chose 7 as their favourite or lucky number? _____
 - How many students chose 8 as their favourite or lucky number? _____
 - How many students chose 4 as their favourite or lucky number? _____
 - Which number was the mode (the number chosen most)? _____
 - Which number was chosen least? _____

- 3 Draw a **frequency histogram** to show this data.

To draw a frequency histogram

- Choose a title for the graph.
- Choose a width for each column. Label the horizontal axis.
- Remember that there are no gaps between the columns.
- Choose a scale and label the vertical axis.
- The columns are centred on each score.
- The first column begins half a column width from the vertical axis.

It may be easier to use a sheet of grid paper for your graph.

6.4 Family pets

- 1 Organise the data from your survey in this table.



Family pets

Type of pet	Tally	Frequency
Dog		
Cat		
Bird		
Fish		
Others		
None		

- 2
- How many pets were counted in the survey? _____
 - Which type of pet was most common? _____
 - What percentage of the pets were dogs? _____
 - What percentage of the pets were cats? _____
 - How many students did not have a pet? _____

- 3 Draw a **column graph** to show this data.

To draw a column graph

- Choose a title.
- Choose a scale and label the vertical axis.
- Choose a width for each column. Label the horizontal axis.
- Draw the graph. The space between each column must be the same.

It may be easier to use a sheet of grid paper for your graph.



6.5 People at home

- 1 Organise the data from your survey in this table.



Number of people living at home

Number of people	Tally	Frequency
1		
2		
3		
4		
5		
6		
7		
8		

- 2
- How many homes had exactly 4 people? _____
 - How many homes had more than 5 people? _____
 - How many homes had less than 4 people? _____
 - What number of people was the mode (most common)? _____
 - Calculate the mean (average) number of people per home. _____

- 3 Draw a **frequency histogram** to show this data.

To draw a frequency histogram

- Choose a title for the graph.
- Choose a width for each column. Label the horizontal axis.
- Remember that there are no gaps between the columns.
- Choose a scale and label the vertical axis.
- The columns are centred on each score.
- The first column begins half a column width from the vertical axis.

It may be easier to use a sheet of grid paper for your graph.

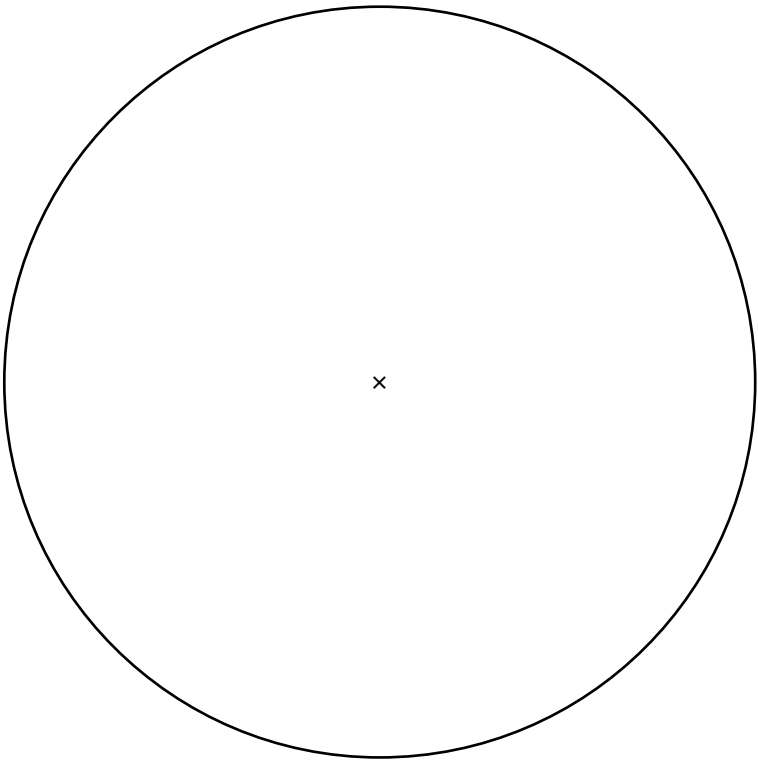
6.6 Favourite sport

- 1 Organise the data from your survey in this table.
(You may wish to analyse the data for boys and girls separately and then compare them. If so, use two copies of this activity sheet.)

Favourite sport		
Sport	Tally	Frequency

- 2 a How many students voted in this survey? _____
b Which sport received the most votes? _____
c Which sport received the smallest number of votes? _____
- 3 Draw a **sector graph** to show this data.

To draw a sector graph, you need to calculate the size of the angle in each sector.
To do this, divide each frequency by the total number of votes, then multiply by 360.



6.7 Household chores

- 1** Organise the data from your survey in this table.
(You may wish to analyse the data for boys and girls separately and then compare them. If so, use two copies of this activity sheet.)

Hours spent on household chores

Number of hours	Tally	Frequency
0		
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		
14		

- 2 a** How many students were counted in this survey?

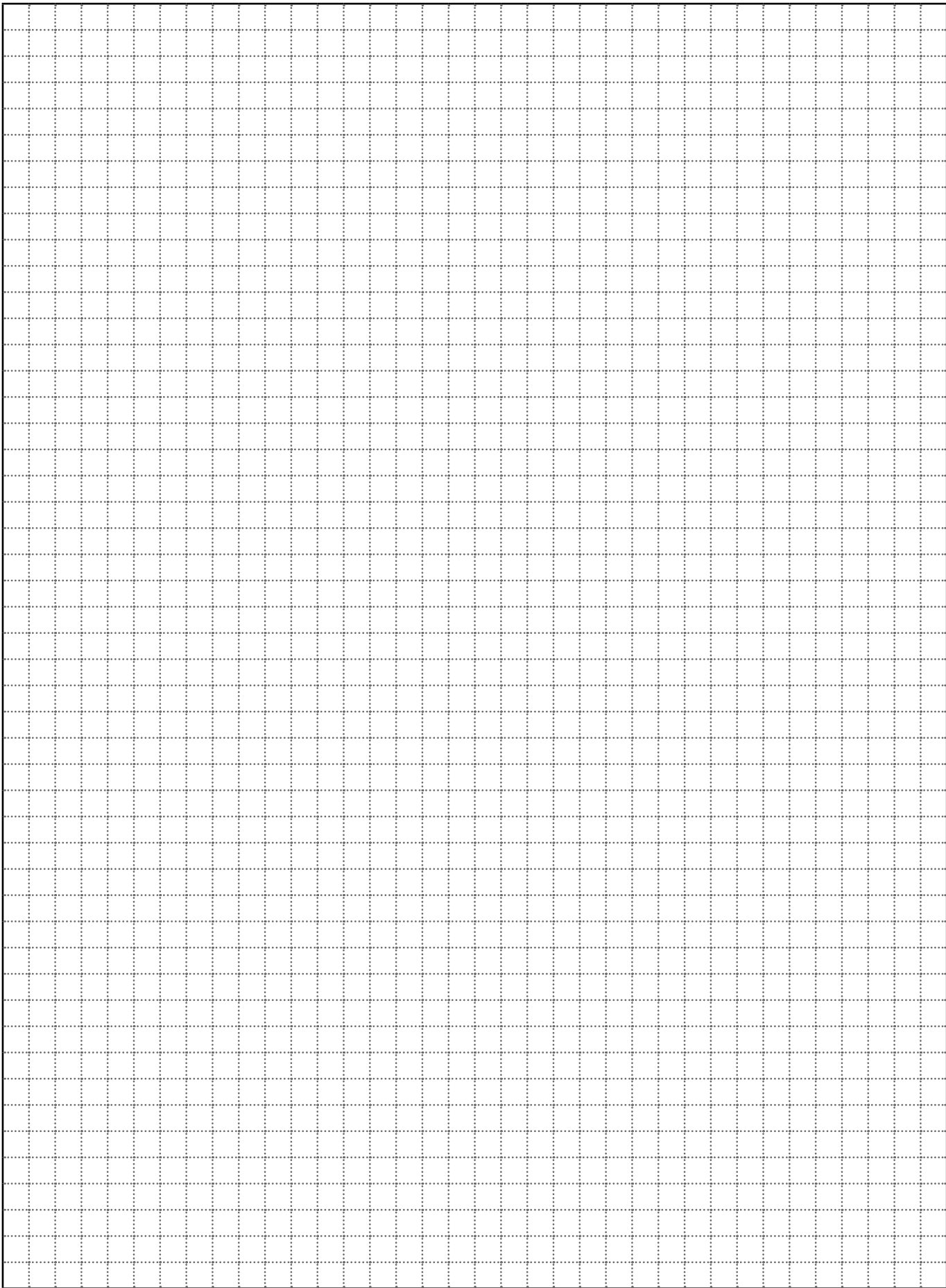
- b** How many students did no household chores last week?

- c** How many students spent 10 hours or more on chores?

- d** How many students spent 5 hours or less on chores?

- e** How many students spent between 5 and 10 hours on chores?

- f** What percentage of students spent **more than 5 hours** on chores? _____
- g** What was the most common amount of time spent on chores? _____
- h** What was the mean (average) amount of time spent on chores? _____
- 3 a** Draw **frequency histogram** to show this data.
- b** Add a **frequency polygon** to your graph.
It may be easier to use a sheet of grid paper for your graphs.



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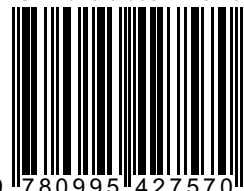
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Instant Lessons

An imprint of Blake Education Pty Ltd
ABN 50 074 266 023
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ISBN 978-0-99542-757-0



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